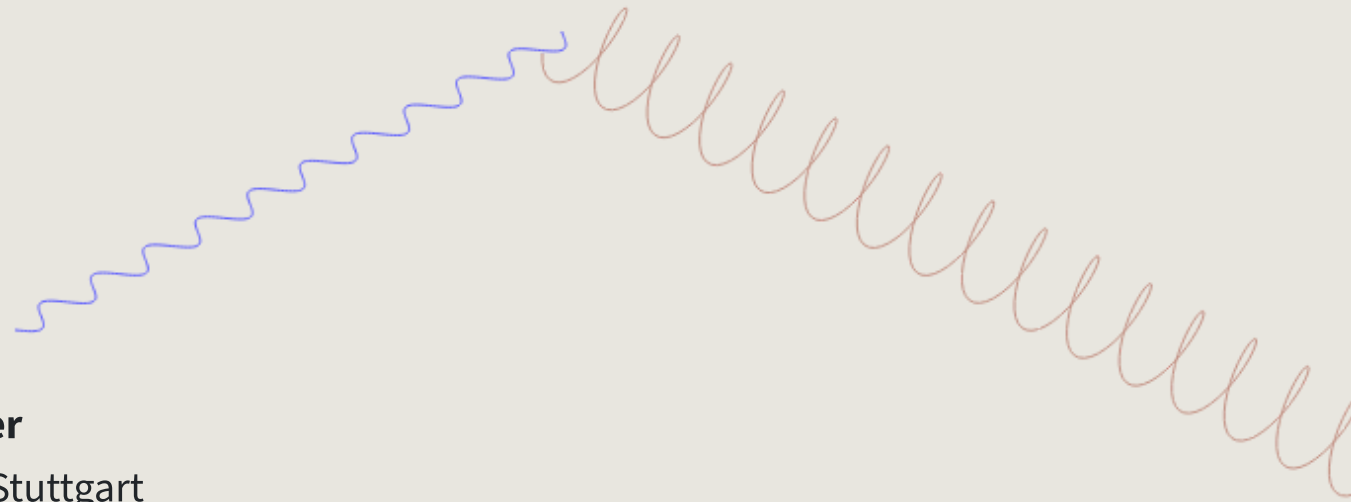


Optical & Magneto-Optical Signatures of Altermagnetism

PROBING THE EFFECT OF SPIN-SPLITTING



Jakob Wallner

University of Stuttgart

[Link to the presentation](#)

Hauptseminar: Talk 8 of 12

27.05.2026



University of Stuttgart
Germany

How do we identify Altermagnets?

The Identification Challenge

Initial Claims (Surface)

- ARPES observed apparent spin-splitting in RuO_2 .
- Led to widespread classification as a model altermagnet.

The Debunking (Bulk)

- Optical spectroscopy revealed **Fermi-liquid behavior**.
- Matches a non-magnetic model better than an altermagnetic one.

Altermagnetism in RuO_2 thin films: A new magnetic material for the AI era

by National Institute for Materials Science

Article | [Open access](#) | Published: 24 September 2025

Evidence for single variant in altermagnetic RuO_2 (101) thin films

[Cong He](#), [Zhenchao Wen](#), [Jun Okabayashi](#), [Yoshio Miura](#), [Tianyi Ma](#), [Tadakatsu Ohkubo](#), [Takeshi Seki](#), [Hiroaki Sukegawa](#) & [Seiji Mitani](#)

VS



Fermi-liquid behavior of nonaltermagnetic RuO_2

Maxim Wenzel,¹ Ece Uykur,^{2,*} Sahana Röbler,³ Marcus Schmidt,³ Oleg Janson,⁴ Achyut Tiwari,¹ Martin Dressel,¹ and Alexander A. Tsirlin^{5,†}

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²*Helmholtz-Zentrum Dresden-Rossendorf, Institute of Ion Beam Physics and Materials Research, D-01328 Dresden, Germany*

³*Max Planck Institute for Chemical Physics of Solids, D-01067 Dresden, Germany*

⁴*Institute for Theoretical Solid State Physics, Leibniz IFW Dresden, D-01069 Dresden, Germany*

⁵*Felix Bloch Institute for Solid-State Physics, University of Leipzig, D-04103 Leipzig, Germany*

Revisiting altermagnetism in RuO_2 : a study of laser-pulse induced charge dynamics by time-domain terahertz spectroscopy

 Check for updates

David T. Plouff,¹ Laura Scheuer,¹ Shreya Shrestha,¹ Welpeng Wu,¹ Nawsher J. Parvez,¹ Subhash Bhatt,¹ Xinhao Wang,¹ Lars Gundlach,² M. Benjamin Jungfleisch,³ & John Q. Xiao¹

Ambiguity remains: Surface effect or Bulk property?
We need a **definitive Bulk Fingerprint**.

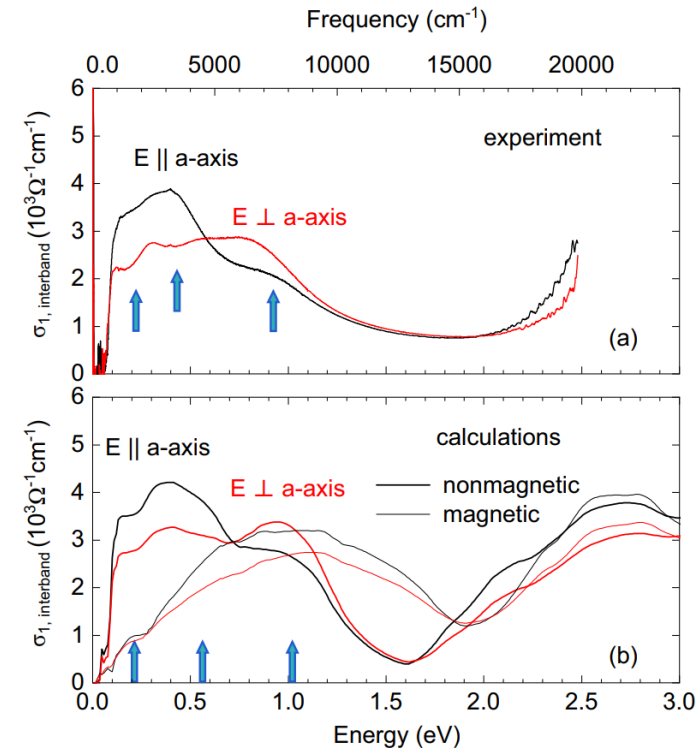
Fermi-liquid behavior of nonaltermagnetic RuO_2

NON-MAGNETIC SIMULATION MATCHES THE EXPERIMENTAL DATA

DFT: Magnetic/Non-Magnetic vs Experiment

"Adding magnetism clearly deteriorates the description of the experimental optical conductivity.

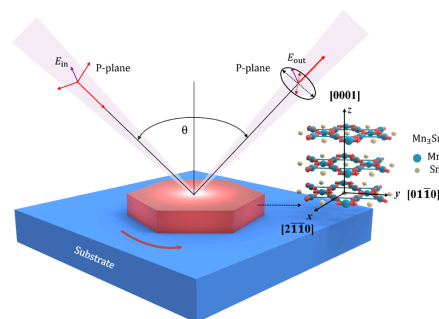
Note in particular the acute discrepancies at low energies where the sharp increase in the optical conductivity (Pauli edge) cannot be reproduced by the magnetic solution."



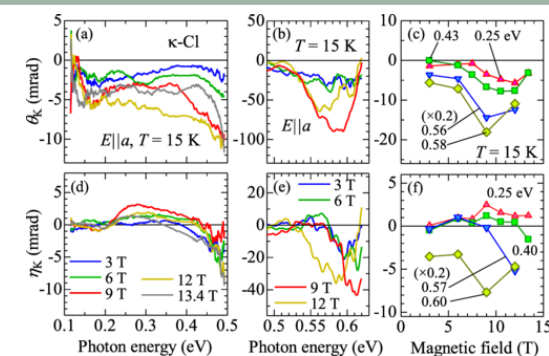
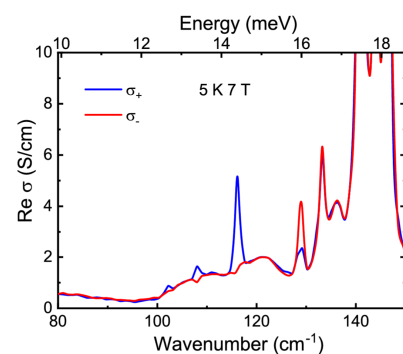
ARPES predicts Altermagnetism
Magneto-Optical measurements show otherwise!

*M. Wenzel et al., Phys. Rev. B **111**, (2025)*

Background: Magneto Optical Kerr Effect & Mueller Matrix



$$S_{\text{out}}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

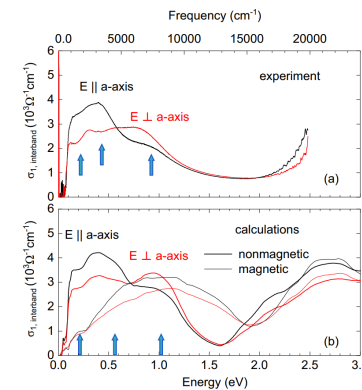
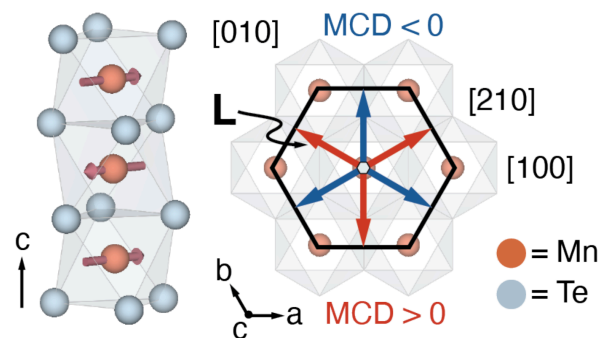


D. Gao et al., APL Mater. **12**, 091114 (2024)
K. Panda et al., arXiv:2507.17238 [cond-mat.str-el] (2025)
S. Iguchi et al., Phys. Rev. Res. **7**, (2025)

Case Studies: Off-Diagonal Elements of $\tilde{\sigma}_{ij}$

$$\tilde{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ -\sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & \sigma_{zz} \end{pmatrix}$$

MnTe

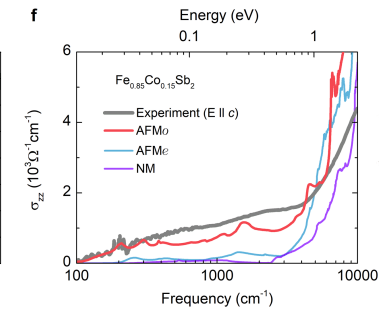
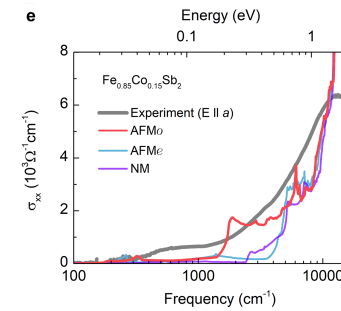
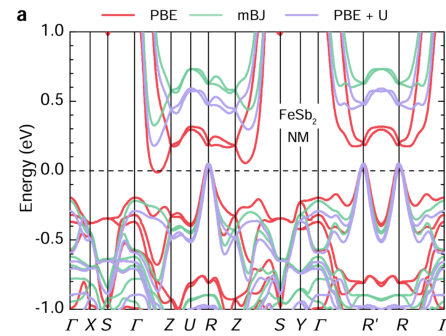


D. Gao et al., APL Mater. **12**, 091114 (2024)
M. Wenzel et al., Phys. Rev. B **111**, (2025)

Case Studies: Diagonal Elements of $\tilde{\sigma}_{ij}$

$$\tilde{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & 0 \\ -\sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & \sigma_{zz} \end{pmatrix}$$

FeSb₂

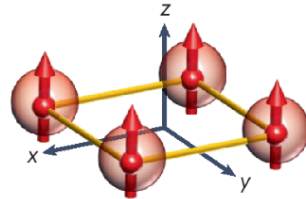


Breaking the Symmetry

TRANSFORMATION SYMMETRY IN DIFFERENT MAGNETIC ORDERS

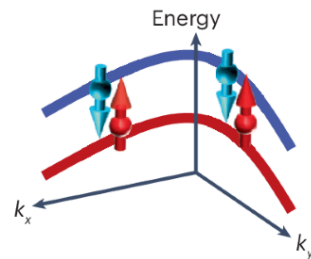
A: FERRO

- $M \neq 0$
- **Breaks** time-reversal symmetry $\mathcal{T}$
- No spin degeneracy



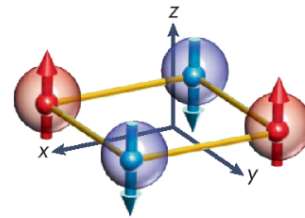
Momentum space

Unidirectional splitting

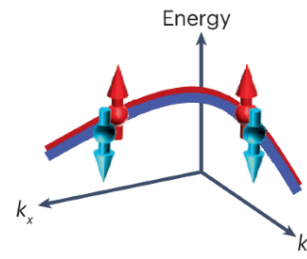


B: ANTI-FERRO

- $M = 0$
- **Preserves** symmetry $\mathcal{PT}$
- Spin-degenerate electronic bands

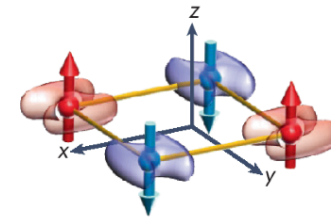


Non-splitting

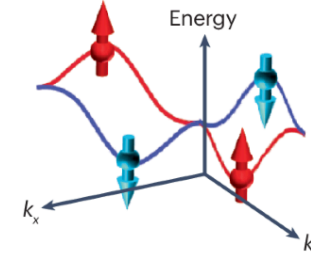


C: ALTERMAGNET

- $M = 0$
- **Breaks** $\mathcal{T}$ but **preserves** $\mathcal{PT}$
- No spin degeneracy



Alternating splitting



Optical Behavior

LINEAR RESPONSE THEOREM AND THE OPTICAL CONDUCTIVITY TENSOR

Normal materials follow **Linear Response Theorem:**

$$J = \sigma E \quad \tilde{\epsilon} \leftrightarrow \tilde{\sigma} \begin{cases} \tilde{\epsilon}_{xx} = 1 + \frac{i\tilde{\sigma}_{xx}}{\epsilon_0\omega} \\ \tilde{\epsilon}_{xy} = \frac{i\tilde{\sigma}_{xy}}{\epsilon_0\omega} \end{cases}$$

$\tilde{\sigma}_{ij}(\omega)$: complex optical conductivity tensor

- **Absolute Reflectivity** $R(\omega)$
- **Rotation** $\Delta\Psi(\omega)$, **Ellipticity** η

⚠️ Problem:

Isolate Altermagnetism as source of off-diagonal elements

ORDINARY:

$$\tilde{\sigma} = \begin{pmatrix} \sigma_{xx} & 0 & 0 \\ 0 & \sigma_{xx} & 0 \\ 0 & 0 & \sigma_{xx} \end{pmatrix}$$

MAGNETIC:

$$\tilde{\sigma} = \begin{pmatrix} \tilde{\sigma}_{xx} & \tilde{\sigma}_{xy} & 0 \\ -\tilde{\sigma}_{xy} & \tilde{\sigma}_{xx} & 0 \\ 0 & 0 & \tilde{\sigma}_{zz} \end{pmatrix}$$

STRUCTURAL:

$$\tilde{\sigma} = \begin{pmatrix} \tilde{\sigma}_{xx} & \tilde{\sigma}_{xy} & 0 \\ +\tilde{\sigma}_{xy} & \tilde{\sigma}_{xx} & 0 \\ 0 & 0 & \tilde{\sigma}_{zz} \end{pmatrix}$$

Optical Behavior

MECHANISMS GENERATING OFF-DIAGONAL TENSOR ELEMENTS (σ_{xy})

Antisymmetric (

$$\sigma_{xy} = -\sigma_{yx})$$

Requires Time-Reversal Breaking

- **External Magnetic Field ($\mathbf{B} \neq 0$):** Explicitly breaks $\mathcal{T}$ via the Lorentz force, cyclotron motion / Ordinary Hall effect.
- **Ferromagnetism ($\mathbf{M} \neq 0$):** Spontaneous global $\mathcal{T}$ -breaking coupled with Spin-Orbit Interaction
- **Altermagnetism ($\mathbf{M} = 0$):** Spontaneous $\mathcal{T}$ -breaking through alternating crystal symmetries.

Symmetric ($\sigma_{xy} = \sigma_{yx}$)

Structural & Reciprocal Origins

- **Crystal Misalignment:** Choosing a laboratory coordinate framework that is rotated relative to the principal crystallographic axes.
- **Linear Birefringence / Dichroism:** Intrinsic optical anisotropy in lower-symmetry crystal systems (like the orthorhombic structure of $FeSb_2$).
- **Piezomagnetic / Magnetoelectric Effects:** Strain-induced or voltage-induced modifications to the tensor framework

Optical Behavior

STRUCTURAL ORIGIN OF σ_{xy} : LOW-SYMMETRY MONOCLINIC/ORTHORHOMBIC SYSTEMS

Example: Structural Monoclinic Angle Alignment

Internal crystal axes (a, b, c) do not match an orthogonal laboratory coordinate framework (x, y, z):

- **Purely Structural:** Occurs in completely non-magnetic, time-reversal symmetric insulators.
- **The Consequence:** Electric fields applied along the laboratory x -axis naturally induce a polarization current along the laboratory y -axis due to the tilted unit cell shape.
- **Tensor Property:** Enforces a strictly **symmetric** off-diagonal element: $\sigma_{xy} = \sigma_{yx}$.

Zirconium Dioxide ZrO_2

Applying E_x yields a projection along both a and b , mechanically generating an J_y response.

Lattice parameters:

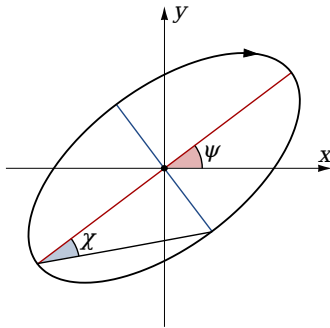
$$[\alpha = 90.00^\circ, \beta = 99.44^\circ, \gamma = 90.00^\circ]$$

The Magneto-Optical Kerr Effect (MOKE)

MEASURING THE OPTICAL CONDUCTIVITY TENSOR

$$\tilde{\sigma}_{xy}(\omega) \propto \theta_K + i\eta_K$$

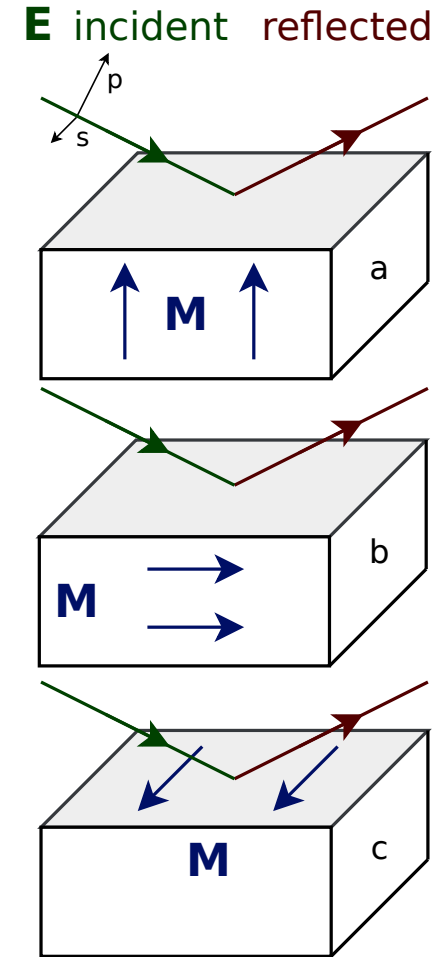
- $\theta_K(\Psi)$ (Rotation) $\leftrightarrow$ Dispersive part, $\text{Im}(\tilde{\sigma}_{xy})$
- η_K (Ellipticity) $\leftrightarrow$ Dissipative part, $\text{Re}(\tilde{\sigma}_{xy})$



Conventions:

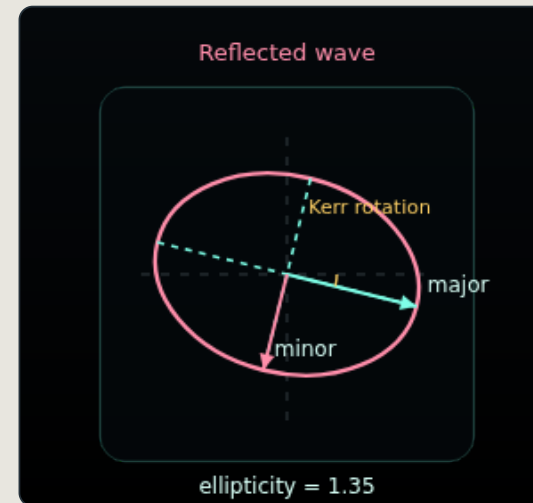
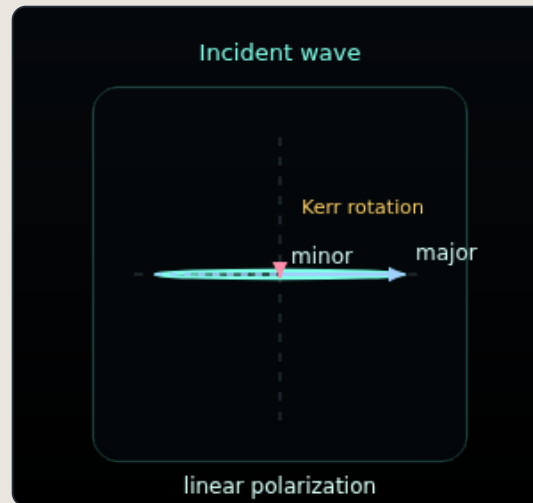
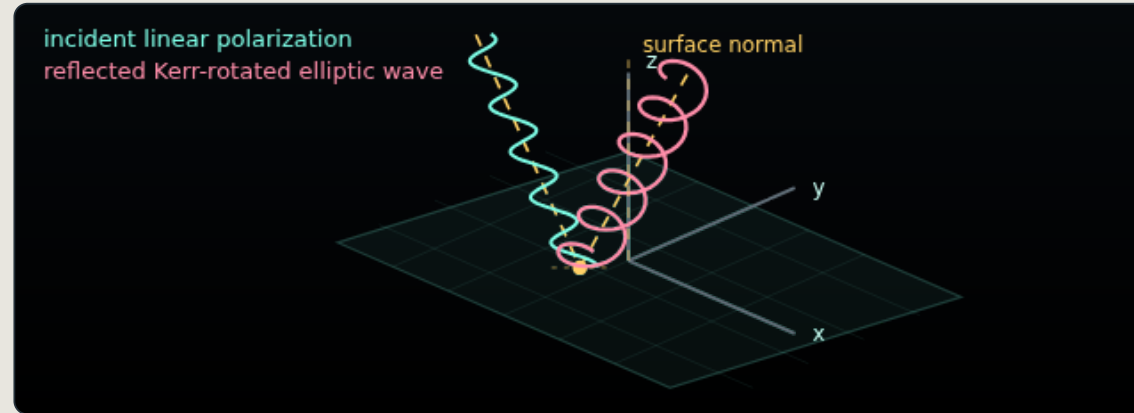
- **a: polar**
- **b: longitudinal**
- **c: transversal**

Polar MOKE: Light interacts only with the xy -plane.
Let's us extract σ_{xy} from the reflected θ_K and η_K .



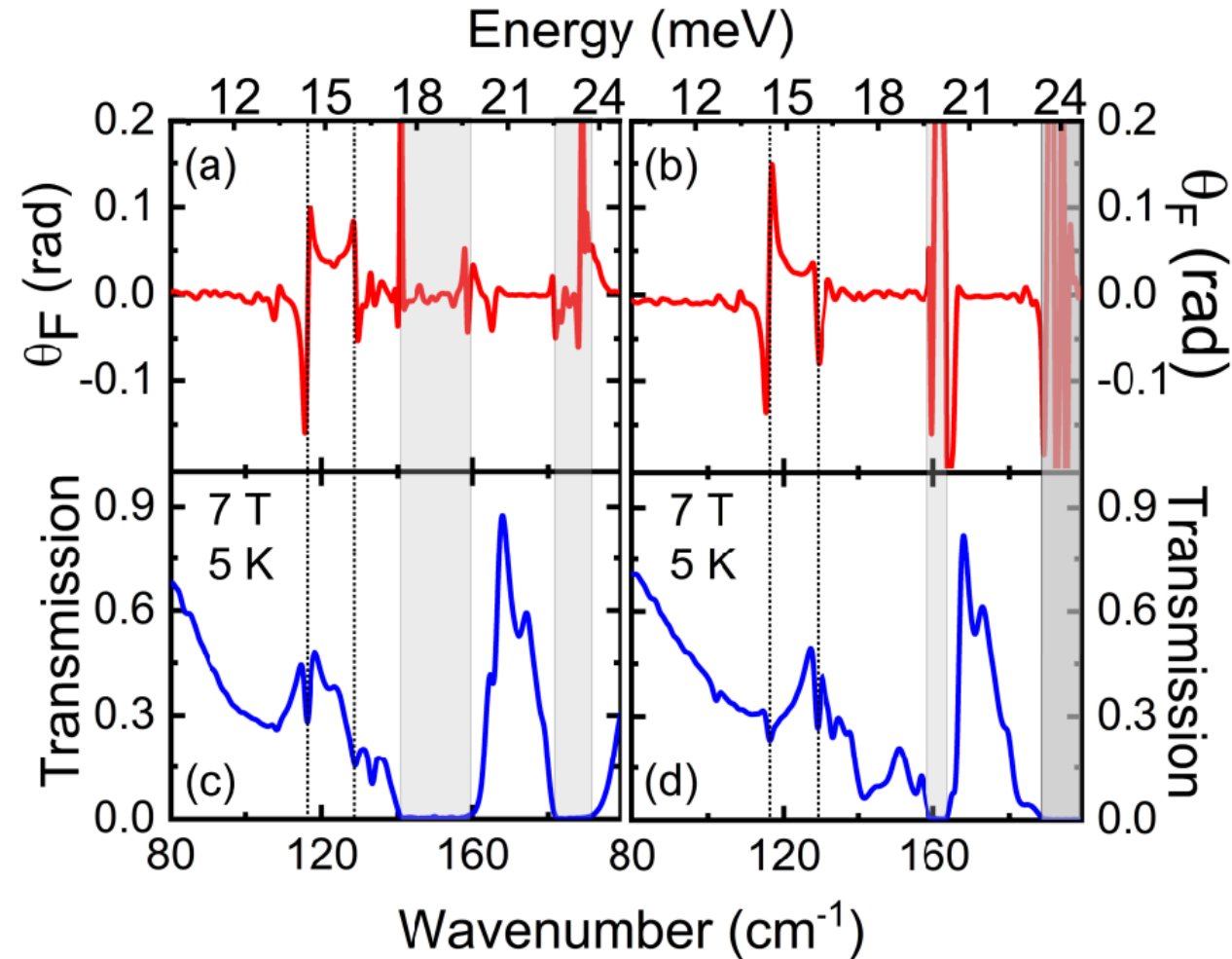
The Magneto-Optical Kerr Effect (MOKE)

MEASURING THE KERR ROTATION AND ELLIPTICITY



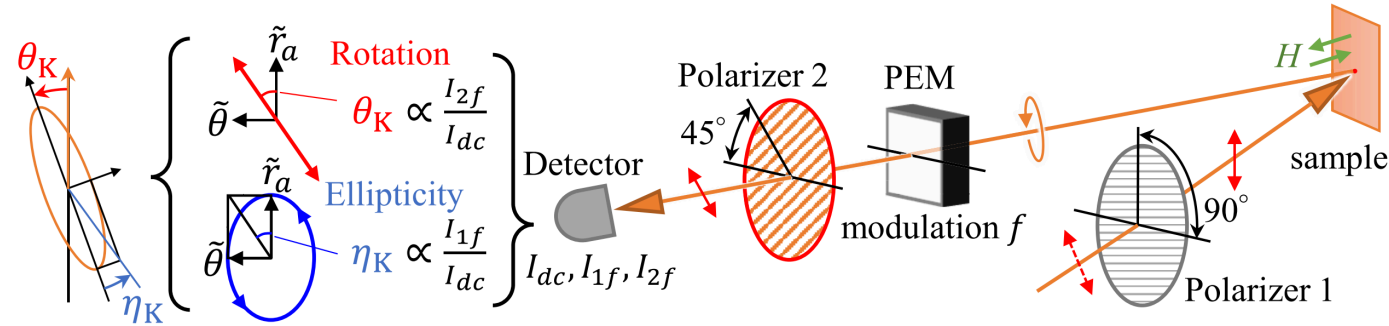
The Magneto-Optical Kerr Effect (MOKE)

MEASURING THE KERR ROTATION AND ELLIPTICITY



Measuring the MOKE Response

EXPERIMENTAL SETUP AND DATA ANALYSIS: SINGLE FREQUENCY



SETUP:

- **Light source:** Monochromatic, linearly polarized light
- **PEM:** Polarization-encoding module
- **Analyzer:** Rotating polarizer
- **Detector:** Intensity measurement device
- **Lock-in Amplifier:** Optional, Signal processing unit

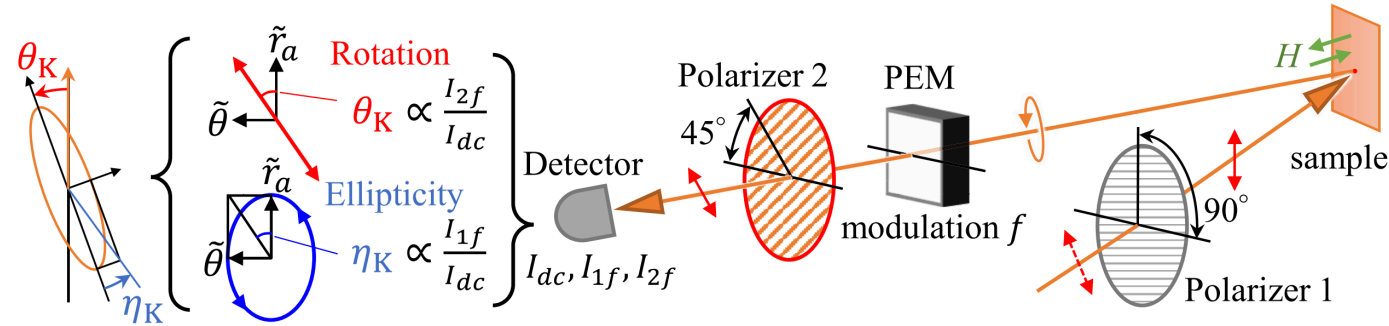
DATA ANALYSIS:

- I_{dc} : Direct current, Absolute Reflectivity $R(\omega)$
- $I_{1\omega}$: First Harmonic, Kerr ellipticity η_K
- $I_{2\omega}$: Second Harmonic, Kerr rotation θ_K

The PEM (Photoelastic Modulator) is a critical component used to highly sensitively detect extremely small rotation angles and ellipticities of the reflected light ($\sim 10^{-3}$ to 10^{-5} radians)

Measuring the MOKE Response

EXPERIMENTAL SETUP AND DATA ANALYSIS: SINGLE FREQUENCY



ADVANTAGES:

- **Sensitivity:** High, up to milliradians
- **Noise:** Low, because of Lock-In Amplifier

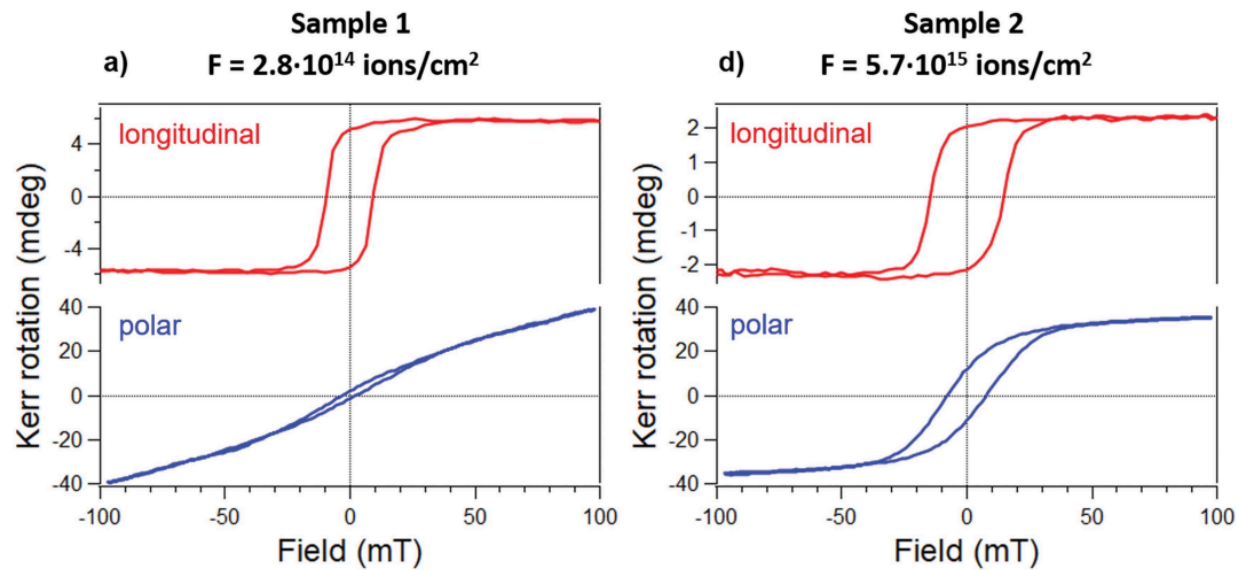
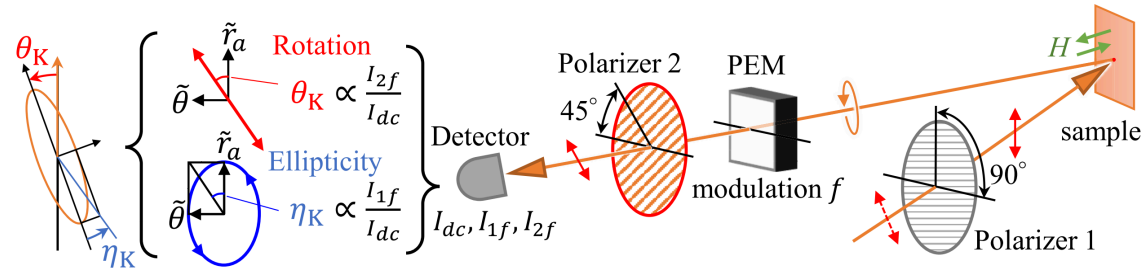
DISADVANTAGES:

- **Chromatic hardware:** Wavelength dependent (e.g. PEM)
- **Resolution:** Low, maybe one frequency

Useless or very high effort for broad band measuring!

Measuring the MOKE Response

EXPERIMENTAL SETUP AND DATA ANALYSIS: SINGLE FREQUENCY



Optical Behavior

LINEAR RESPONSE THEOREM

CONCEPT

For broadband spectrum, **remove all wavelength-dependent hardware** (like waveplates and PEMs), use wideband linear polarizers.

PROBLEM

Linear polarizers can measure absolute reflectivity R and Kerr rotation θ_K but without a waveplate, **we cannot measure ellipticity!**

Optical Behavior

LINEAR RESPONSE THEOREM

✎ CONCEPT

Remove all wavelength-dependent hardware

⚠ PROBLEM

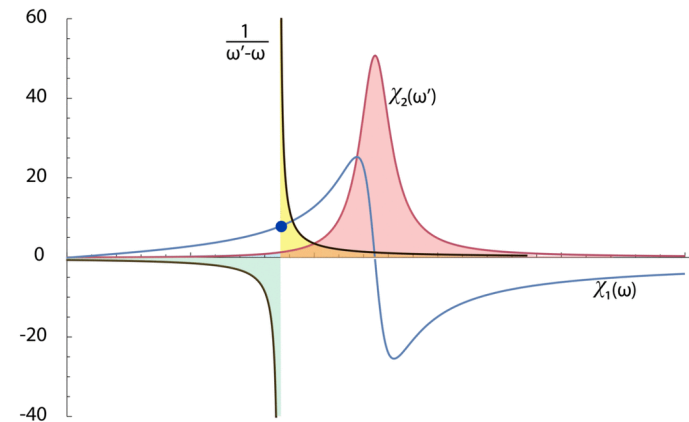
We cannot measure ellipticity!

KRAMERS KRONIG RELATION

Condition: Response function (analytic in the upper half-plane).

$$\operatorname{Re}[f(\omega)] = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\operatorname{Im}[f(x)]}{x - \omega} dx$$

$$\operatorname{Im}[f(\omega)] = -\frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{\operatorname{Re}[f(x)]}{x - \omega} dx$$



<https://upload.wikimedia.org/wikipedia/commons/4/45/Kramersgifwiki.gif>
(2026)

Magneto-Optical Kramers-Kronig Analysis

EXPERIMENTAL SETUP AND DATA ANALYSIS: BROAD BAND

Workflow:

Measuring:
 $R(\omega)$ & $\theta_K(\omega)$



KK Transform:
 $f(\omega) = \ln c(\omega) + i \cdot 2\theta_K(\omega)$



Extract:
 $R_+(\omega)$ & $R_-(\omega)$



Final KK:
 $\phi_+(\omega)$ & $\phi_-(\omega)$



Full Complex
 $\tilde{r}_\pm(\omega) = \sqrt{R_\pm} e^{i\phi_\pm}$

Mathematical Engine:

- Causal Auxiliary Function:

$$f(\omega) = \ln \left(\frac{\tilde{r}_-}{\tilde{r}_+} \right) = \ln c(\omega) + i \cdot 2\theta_K(\omega)$$

- 1st KK Transform (Yields Dichroism):

$$\ln c(\omega) = \frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{2\theta_K(x)}{x^2 - \omega^2} dx$$

- Algebraic Separation:

$$R = \frac{R_+ + R_-}{2} \quad \text{and} \quad c = \frac{R_-}{R_+}$$

- 2nd KK Transform (Yields Phase):

$$\phi_\pm(\omega) = -\frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{\ln \sqrt{R_\pm(x)}}{x^2 - \omega^2} dx$$

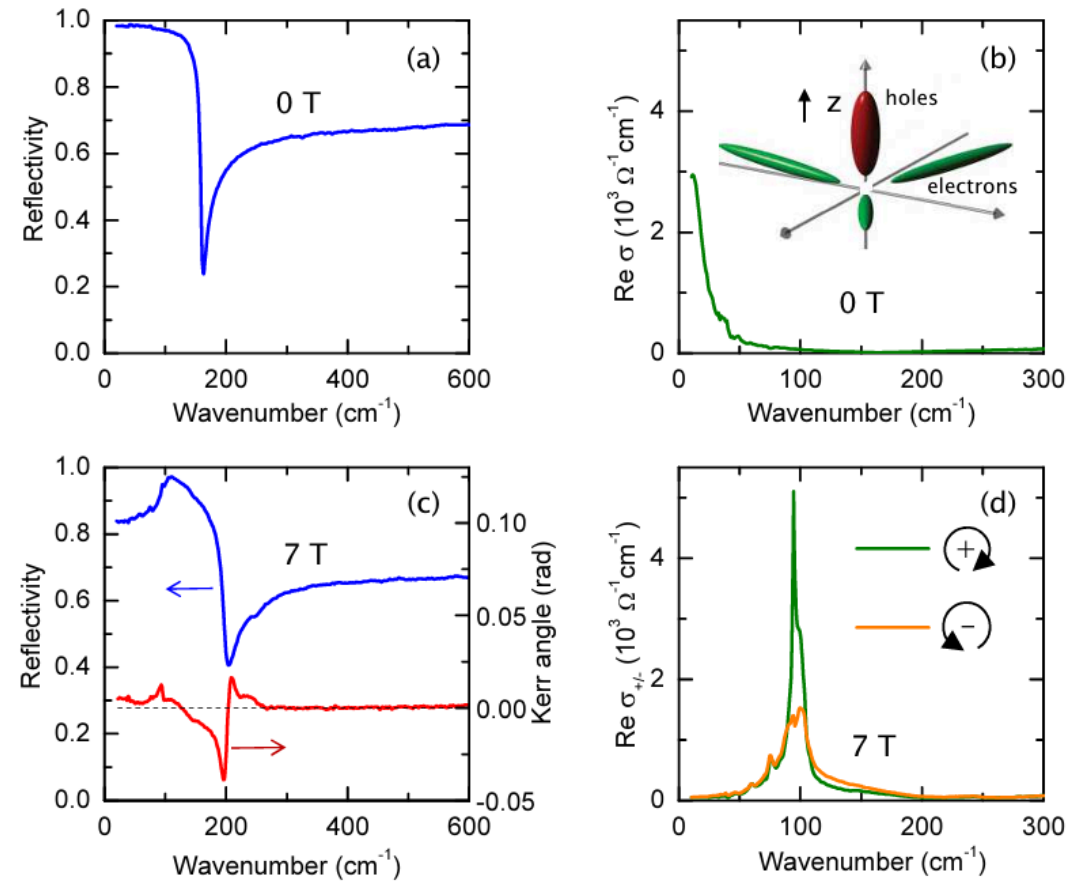
Magneto-Optical Kramers-Kronig Analysis

EXPERIMENTAL SETUP AND DATA ANALYSIS: BROAD BAND

Measurement

Far-infrared optical and magneto-optical spectra of bismuth at 10K.

- (a) and (b): Zero field reflectivity and optical conductivity.
- (c): Magneto-reflectivity (blue) and Kerr rotation (red) at 7T.
- (d): Real part of optical conductivity for the two circular polarizations obtained using MOKKA



Magneto-Optical Kramers-Kronig Analysis

EXPERIMENTAL SETUP AND DATA ANALYSIS: BROAD BAND

- **Auxiliary Function:**

$$f(\omega) = \ln\left(\frac{\tilde{r}_-}{\tilde{r}_+}\right) = \ln c(\omega) + i \cdot 2\theta_K(\omega)$$

- **Kramers-Kronig:**

$$\ln c(\omega) = \frac{2\omega}{\pi} \mathcal{P} \int_0^\infty \frac{2\theta_K(x)}{x^2 - \omega^2} dx$$

- **Reconstruct & Solve:**

$$R = \frac{R_+ + R_-}{2} \quad c = \frac{R_-}{R_+}$$

ADVANTAGE

- Avoids the spectral limitations of wave plates
- Still highly sensitive to small signals
- Ideal for altermagnets

DISADVANTAGE

Measuring in extreme environments?

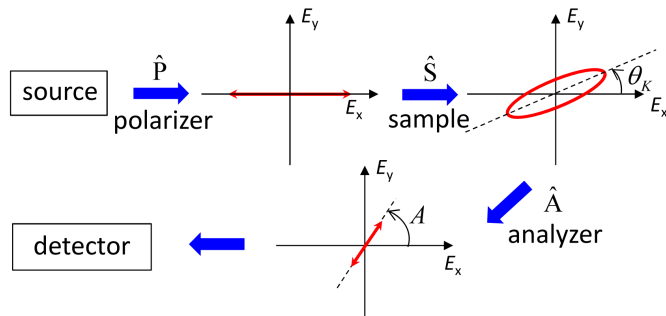
Rotating equipment obstruct high magnetic fields & low temperatures!

MOKKA: The 45-Degree Fast Protocol

BROAD BAND MEASURING IN EXTREME ENVIRONMENTS

Setup

- Polarizer and Analyzer are fixed at 45° to each other
- This geometry **maximizes sensitivity** to rotation



Analysis

- Detector intensity proportional to the Kerr angle, for small angles:

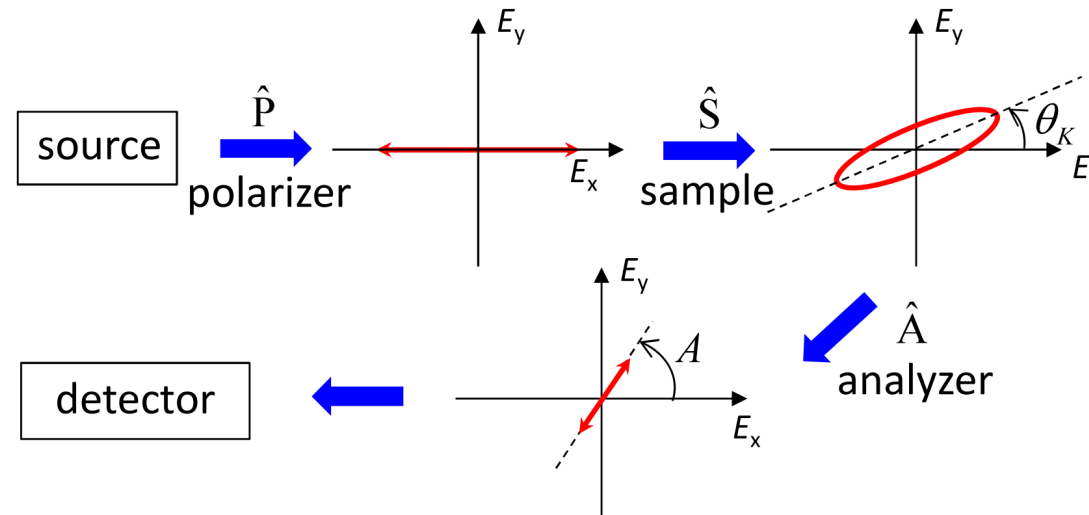
$$I_{\pm 45^\circ} \approx C \cdot R(\omega) \frac{1 \pm 2\theta_K(\omega)}{2}$$

- Measure intensity ratio at positive and negative magnetic fields and isolate θ_K :

$$\rho_{\pm 45^\circ}(B) = \frac{I_{\pm 45^\circ}(+B)}{I_{\pm 45^\circ}(-B)} \approx 1 \pm 4\theta_K(B)$$

MOKKA: The 45-Degree Fast Protocol

BROAD BAND MEASURING IN EXTREME ENVIRONMENTS



Disadvantage

Limited spectral resolution and sensitivity compared to other techniques.

Advantage

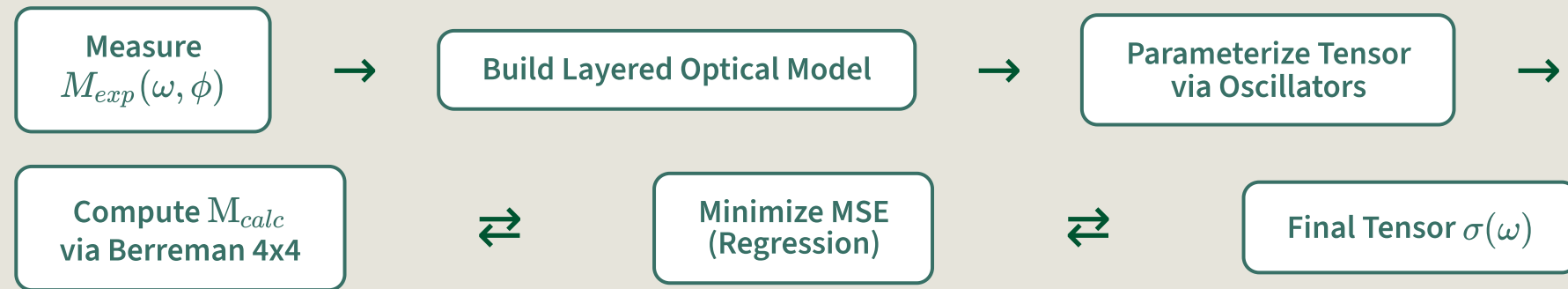
Completely fit for comparing results with DFT calculations

Alternative: Mueller Matrix

A VERY SHORT INTRODUCTION

Mueller Matrix Analysis

ALTERNATIVE OPTICAL RESPONSE DECOMPOSITION



Mueller Matrix Analysis

ALTERNATIVE OPTICAL RESPONSE DECOMPOSITION



Mueller Matrix Formula:

Measure the Mueller Matrix by varying the angle of incidence (Φ) and scanning the photon energy ($\hbar\omega$):

$$S_{\text{out}}(\omega) = M(\omega) \cdot S_{\text{in}}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

- ordinary refractive index and thickness data.
- cross-polarization induced by the off-diagonal elements of your optical conductivity tensor

Mueller Matrix Analysis

ALTERNATIVE OPTICAL RESPONSE DECOMPOSITION



$$S_{\text{out}}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix}$$

- ordinary refractive index and thickness data.
- cross-polarization induced by the off-diagonal elements of your optical conductivity tensor

Geometric Model Construction of the Sample Stack

[Ambient (Air)] → [Surface Roughness] → [Anisotropic/MO Thin Film] → [Substrate]

Typical dielectric tensors for the material layers:

$$\tilde{\epsilon}(\omega) = \begin{pmatrix} \epsilon_{xx}(\omega) & \epsilon_{xy}(\omega) & 0 \\ -\epsilon_{xy}(\omega) & \epsilon_{yy}(\omega) & 0 \\ 0 & 0 & \epsilon_{zz}(\omega) \end{pmatrix}$$

Mueller Matrix Analysis



$$S_{out}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} \quad \tilde{\epsilon}(\omega) = \begin{pmatrix} \epsilon_{xx}(\omega) & \epsilon_{xy}(\omega) & 0 \\ -\epsilon_{xy}(\omega) & \epsilon_{yy}(\omega) & 0 \\ 0 & 0 & \epsilon_{zz}(\omega) \end{pmatrix}$$

Modelling the tensor elements:

- Lorentz oscillators
- Tauc-Lorentz models
(semiconductors/insulators)
- Drude terms (metals)

Fitting Functions:

$$\epsilon_{xy}(\omega) = \sum_j \left[\frac{A_j^+}{\omega_{0,j}^+ - \omega - i\Gamma_j^+} - \frac{A_j^-}{\omega_{0,j}^- - \omega - i\Gamma_j^-} \right]$$

$$\epsilon_{xx}(\omega) = \epsilon_\infty + \sum_k \frac{A_k}{\omega_{0,k}^2 - \omega^2 - i\Gamma_k \omega}$$

System is reduced to a finite set of global fit parameters:

amplitudes (A) , **resonance energies (ω_0)** , and **broadenings (Γ)**

Mueller Matrix Analysis



$$S_{out}(\omega) = \begin{pmatrix} M_{11} & M_{12} & M_{13} & M_{14} \\ M_{21} & M_{22} & M_{23} & M_{24} \\ M_{31} & M_{32} & M_{33} & M_{34} \\ M_{41} & M_{42} & M_{43} & M_{44} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} \quad \tilde{\epsilon}(\omega) = \begin{pmatrix} \epsilon_{xx}(\omega) & \epsilon_{xy}(\omega) & 0 \\ -\epsilon_{xy}(\omega) & \epsilon_{yy}(\omega) & 0 \\ 0 & 0 & \epsilon_{zz}(\omega) \end{pmatrix}$$

$$\epsilon_{xx}(\omega) = \epsilon_{\infty} + \sum_k \frac{A_k}{\omega_{0,k}^2 - \omega^2 - i\Gamma_k\omega} \quad \epsilon_{xy}(\omega) = \sum_j \left[\frac{A_j^+}{\omega_{0,j}^+ - \omega - i\Gamma_j^+} - \frac{A_j^-}{\omega_{0,j}^- - \omega - i\Gamma_j^-} \right]$$

Regression Analysis and Extracting the Conductivity Tensor

- Vary oscillator parameters and layer thicknesses to minimize a global Mean Squared Error (MSE)
- Extract the complex optical conductivity tensor $\tilde{\sigma}(\omega)$ from the dielectric tensor $\tilde{\epsilon}(\omega)$

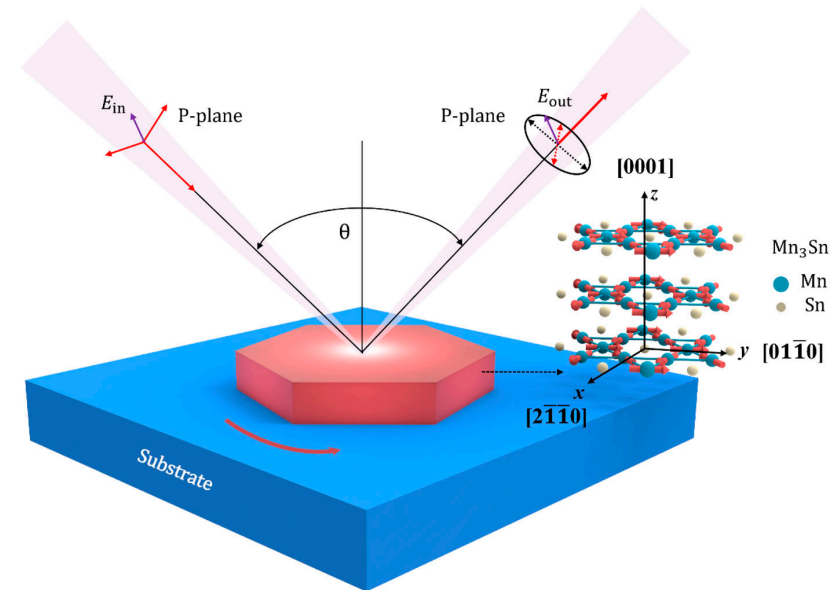
$$\tilde{\sigma}(\omega) = -i\epsilon_0\omega [\tilde{\epsilon}(\omega) - \mathbf{I}]$$

Mueller Matrix Measurement

ANTIFERROMAGNETIC WEYL SEMI-METAL Mn_3Sn

Setup

- **Optical Range:** 300 – 1690 nm (UV to Near-IR).
- **Incident Angles:** 45° – 75°
- **Magnetic Control:** Permanent magnet aligned along specific crystalline orientations.
- **Thermal Control:** Custom semiconductor allows **temperature sweeps** (300 – 400 K).
- **Inversion:** Convert Mueller Matrix into anisotropic permittivity tensor via the **4×4 Transfer Matrix Method**.

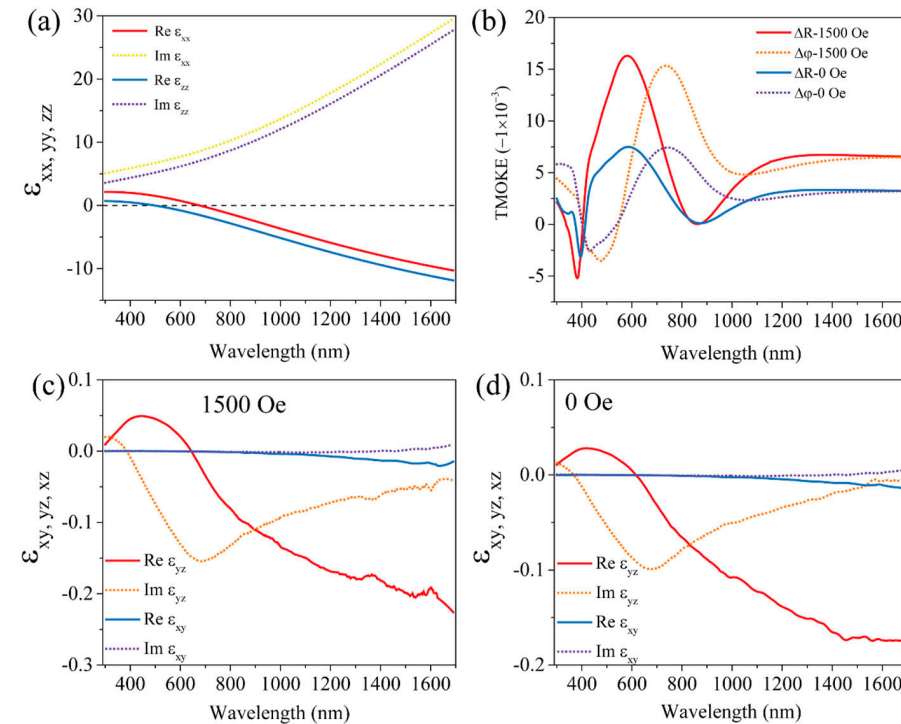


Mueller Matrix Measurement

DIRECT MAPPING OF CRYSTAL ANISOTROPY

Analysis

- **Diagonal Response ($\epsilon_{xx}, \epsilon_{zz}$):** Maps the fundamental, highly anisotropic electronic transitions along different crystal axes
- **Off-Diagonal Response ($\epsilon_{xy}, \epsilon_{yz}$):** Captures the spontaneous magneto-optical activity
- **The Metric:** Evaluated through the Transverse Magneto-Optical Kerr Effect (TMOKE) response across the spectrum.



Comparison between MOKKA and Mueller

FEATURE	MOKKA (DEDICATED MOKE)	MUELLER MATRIX POLARIMETRY
Primary Output	Pure Kerr Rotation & Ellipticity	Full 16 elements describing complete optical transfer function
Acquisition Speed	High Speed / Real-Time Ideal for dynamic, rapid M vs. H loops via PEM/lock-in	Slow / Static States Requires multiple mechanical or synchronized optical modulations
Data Complexity	Low post-processing; immediate physical parameter readout	High; requires algebraic matrix decomposition (e.g., Lu-Chipman)
Anisotropic Crystals (e.g., Altermagnets / RuO ₂)	Vulnerable to artifacts; mixes crystal birefringence with magnetic signal	Flawless Separation Mathematically isolates true magneto-optic cross-coupling
Depolarization	Cannot decouple; registers as loss of signal intensity or baseline shift	Explicitly Quantified Isolates depolarization caused by surface roughness or domains
Core Trade-off: Choose MOKKA for clean, fast magnetic dynamics on isotropic substrates. Choose Mueller Matrix to decouple magnetic ordering from complex, anisotropic crystalline backgrounds.		

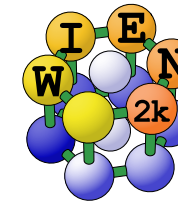
01

Simulation methods

QM SIMULATION AS REFERENCE TO EXPERIMENTAL RESULTS

Optical Response Experiments vs Predictions from DFT Simulations

- Calculate band structure
- Use Kubo formula to calculate optical conductivity tensor
- Compare with experimental MOKE data to confirm altermagnetic order



 VASP

Reminder

*The "correct" choice of exchange correlation functional in DFT is crucial for accuracy for each **individual** system.*

We need a path from band structure to optical response:

Kubo Formula

The Kubo Formula

LINKING BAND STRUCTURE TO OPTICAL RESPONSE

$$\tilde{\sigma}_{xy}(\omega) \propto \frac{1}{\omega} \sum_{n,m} \int_{\text{BZ}} d\mathbf{k} \frac{f(E_{n,\{\mathbf{k}\}}) - f(E_{m,\mathbf{k}})}{E_{m,\mathbf{k}} - E_{n,\mathbf{k}} - \hbar\omega - i\Gamma} \times \langle \psi_{n,\mathbf{k}} | \hat{v}_x | \psi_{m,\mathbf{k}} \rangle \langle \psi_{m,\mathbf{k}} | \hat{v}_y | \psi_{n,\mathbf{k}} \rangle$$

1. The Resonance (Denominator)

- $(E_m - E_n - \hbar\omega)$ dictates **when** a peak occurs
- Vertical interband transitions at specific $\mathbf{k}$ -points

2. The Topology (Matrix Elements)

- $\langle \hat{v}_x \rangle \langle \hat{v}_y \rangle$ contains spin-sign
- In altermagnets, this product survives the BZ sum

◆ Symmetry Note:

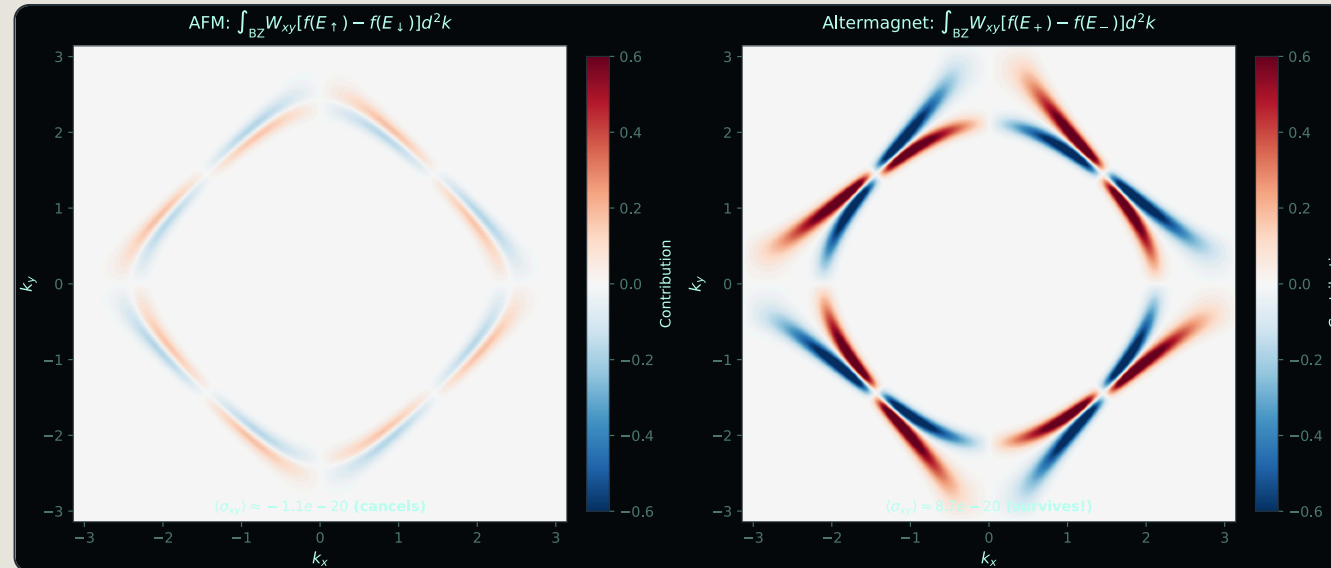
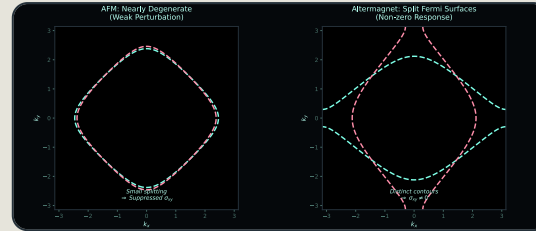
In standard AFMs,
 $\sigma_{xy}(\mathbf{k}) + \sigma_{xy}(\mathcal{PT}\mathbf{k}) = 0.$

◆ Altermagnetism:

Alternating spin-polarization in $\mathbf{k}$ -space $\rightarrow$ net non-zero sum for the off-diagonal elements.

The Kubo Formula

$$\tilde{\sigma}_{xy}(\omega) \propto \dots \sum \int \frac{\Delta f}{\Delta E - \hbar\omega} \times \langle \hat{v}_x \rangle \langle \hat{v}_y \rangle$$



Standard AFM: $\sigma_{xy} = 0$

Altermagnet: $\sigma_{xy} \neq 0$

Off-Diagonal Tensor Elements

Strain and Temperature Restrictions for MnTe

The Candidate: Hexagonal α – MnTe

SYMMETRY PROTECTION & ORDER PARAMETER

THE ORDER PARAMETER

Néel Vector $\mathbf{L}$:

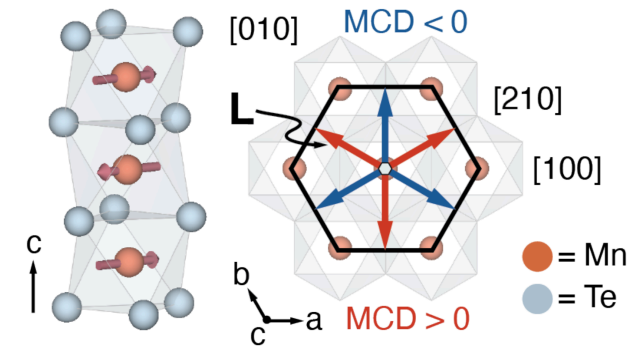
$$\mathbf{L} = \mathbf{M}_1 - \mathbf{M}_2$$

Probed via localized Mn²⁺ ions hosting high-spin $S = 5/2$, $L = 0$ states.

SYMMETRY & TOPOLOGY

- Two magnetic sublattices mapped onto each other via a **six-fold screw rotation**.
- Broken time-reversal symmetry ($\mathcal{T}$) combined with spatial translation enforces a **d-wave altermagnetic state**.
- The crystal hosts a native **six-fold magnetocrystalline anisotropy (MCA)** pinning landscape.

Structure of α – MnTe



Challenge for Production

To make altermagnets industrially viable for spintronics, we need large, macroscopically uniform domains. Knowing how external and internal stress fields alter this magnetic ordering dictates exactly how these devices must be fabricated and handled.

Coherent Néel Vector Steering

MAGNETO-OPTICAL MEASUREMENTS OF MCD, ϕ_0 AND Δ

Measurements

ϕ_0 the birefringence angle with respect to the $[100]$ direction.

Calculations via reflectivity matrix $\tilde{r}$ with
 $r_0 = (r_{xx} + r_{yy})/2$.

$\text{MCD} = \text{Im}[(r_{xy} - r_{yx})/2r_0] \propto L^3 \sin(3\theta_L)$
the magneto-circular dichroism

θ_L the Neel vector orientation

ϕ_0 the birefringence angle

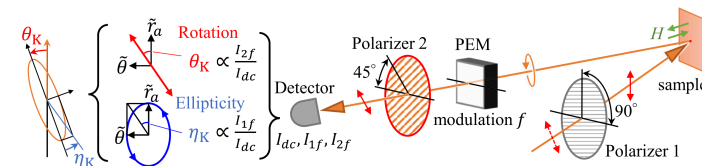
$\Delta = \text{Re}[(r_{xx} - r_{yy})/2r_0]$ the birefringence amplitude

Connection to MOKE

The measurements are closely related to the optical conductivity tensor $\sigma_{ij}(\omega)$, and the dielectric tensor $\epsilon_{ij}(\omega)$. The measurements relate through the intensities measured by MOKE.

$$I_\omega \propto \text{MCD}$$

$$I_{2\omega}(\phi) \propto \Delta \cdot \cos 2(\phi + \phi_0)$$

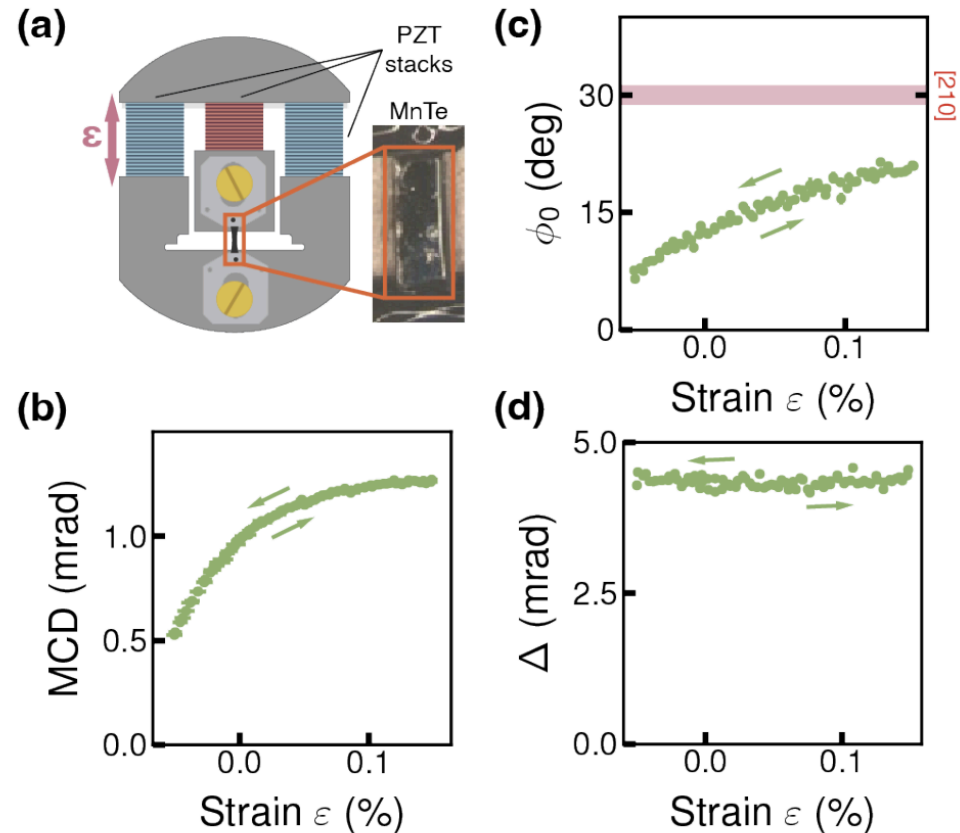


Coherent Néel Vector Steering

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN

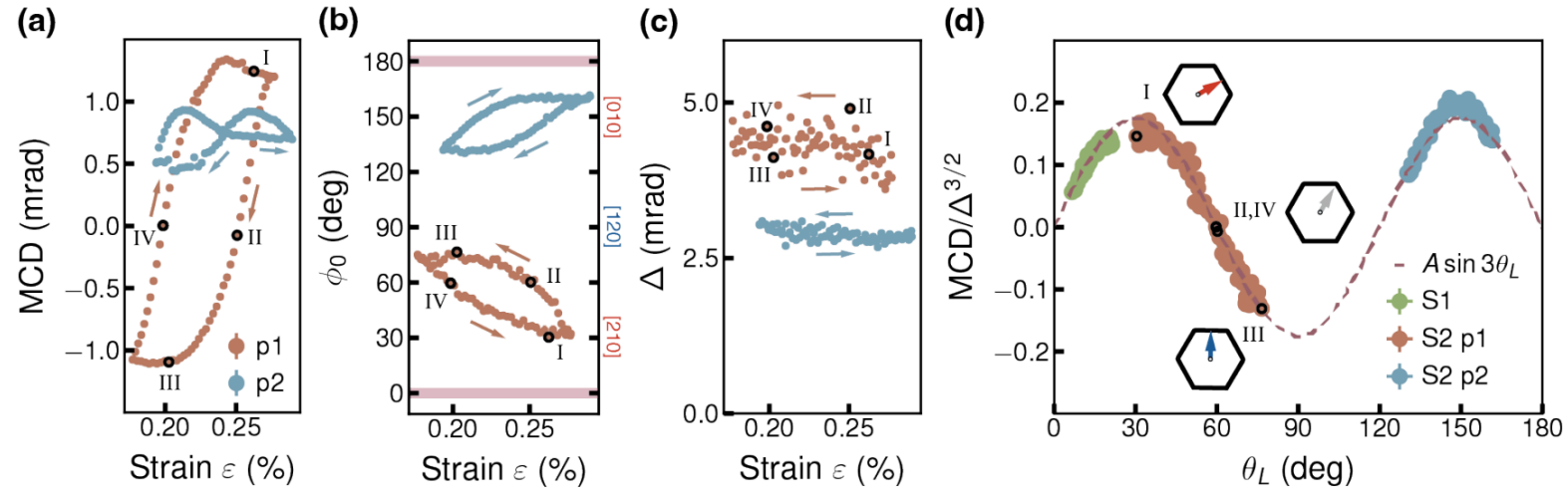
Strain Dependence

- MCD and ϕ_0 show hysteresis like behavior
- Birefringence Amplitude Δ is nearly constant over different strain
- All measurements performed at $\omega/2\pi \approx 50$ kHz



Coherent Néel Vector Steering

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN



Strain Dependence

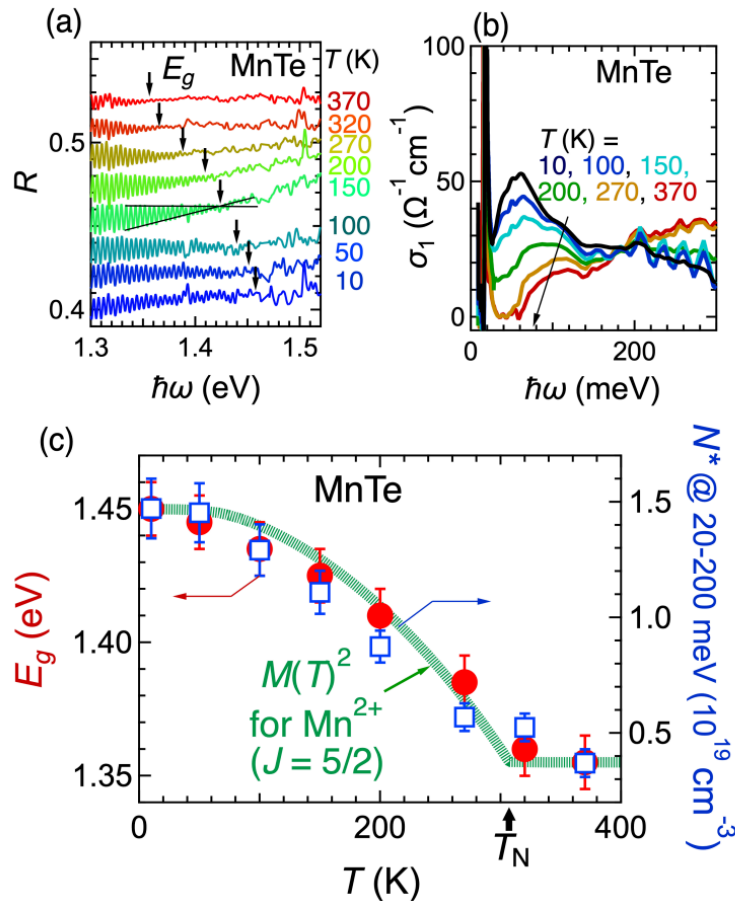
- MCD and ϕ_0 show hysteresis like behavior
- Birefringence Amplitude Δ is nearly constant over different strain

Néel Vector Orientation

- MCD fits to $\sin(\theta_L)$
- Néel Vector is the primary order parameter

Temperature and Strain dependence of MnTe

OPTICAL MEASUREMENTS AS INDICATOR OF ALTERMAGNETISM



Observations

Figure (a) shows the temperature dependence of the interference fringes near the energy gap. The fringes arise due to the transparent character of the thin sample.

- Figure (b): Pure optical conductivity measurement
- Figure (c): Clear $E_g \propto M(T)^2$ dependence

Temperature dependence of E_g in MnTe with **antiferromagnetic spin** arrangement and **no net magnetism** :

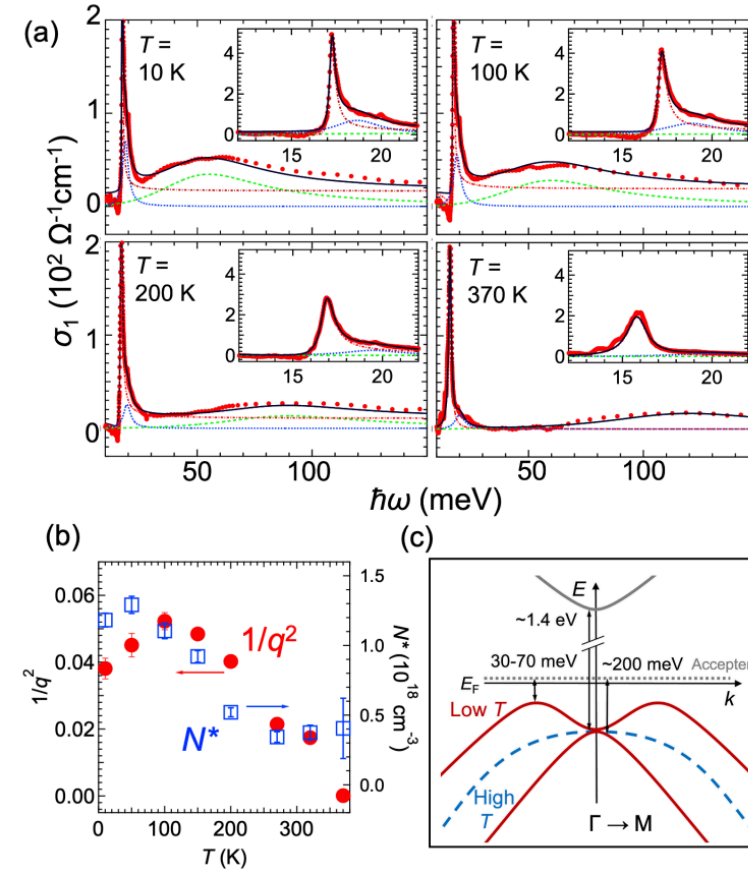
Material might be altermagnet with band splitting like ferromagnetic materials

Temperature and Strain dependence of MnTe

OPTICAL MEASUREMENTS AS INDICATOR OF ALTERMAGNETISM

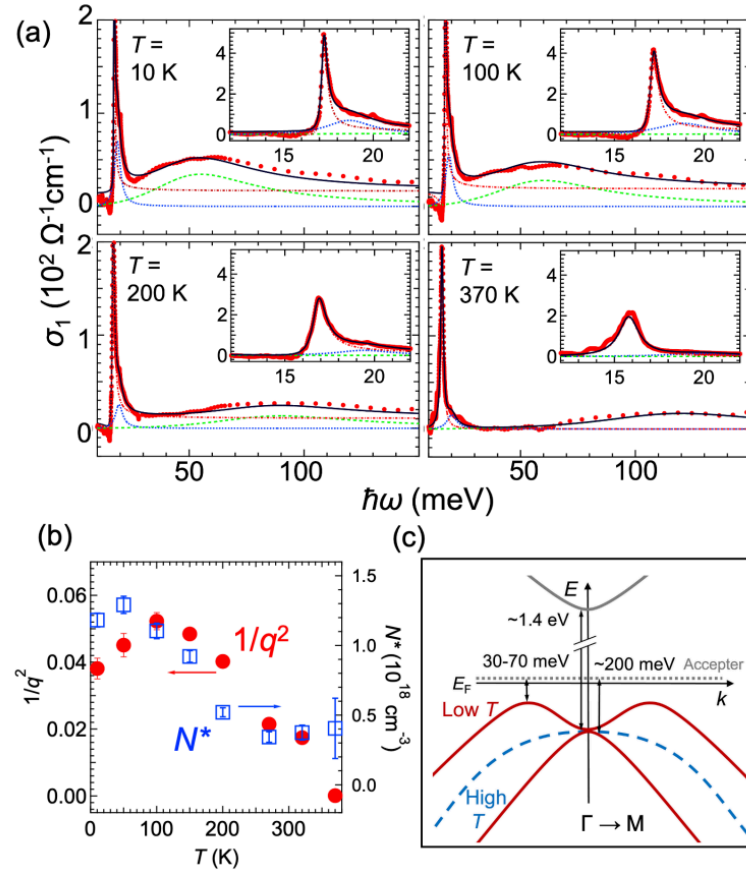
Observations

- Figure (a): $\sigma_1(\omega)$ spectra (circles) at $T = [10, 100, 200, 370]$ K for $\hbar\omega < 150$ meV.
- Figure (b): Fano asymmetric parameter ($1/q^2$, red solid circles) of phonon peak and peak intensity (N^* , blue open squares) of the 20-meV peak as a function of T .
- Figure (c): Expected temperature-dependent electronic-structure change near E_F and the origins of the peaks in $\sigma_1(\omega)$ spectra.



Temperature and Strain dependence of MnTe

OPTICAL MEASUREMENTS AS INDICATOR OF ALTERMAGNETISM



Assumptions

- **Temperature dependent Fano Term** from in-gap states which interact with phonons at low temperatures.
- The in-gap state is thought to originate from the altermagnetic spin-split bands.
- The altermagnetic electronic structure appearance influences the Mn-Te stretching mode.

Conclusion for MnTe

PROBING OFF-DIAGONAL ELEMENTS VIA IN-SITU STRAIN

Magneto-Optical Trackers

By mapping the complex reflection coefficients, isolate elements linked directly to the off-diagonal dielectric tensor $\epsilon_{xy}(\omega)$:

- $I_\omega \propto \text{MCD} \propto L^3 \sin(3\theta_L)$
- $I_{2\omega}(\phi) \propto \Delta \cdot \cos[2(\phi + \phi_0)]$
- $\Delta \propto L^2$

The Discovery

Under applied stress, **MCD and ϕ_0 evolve continuously**, while **Δ remains constant**.
The $1/q^2$ **Fano parameter scaling** supports spin-splitting.

Conclusion

The stress is not changing the size of the altermagnetic order parameter (L). It is continuously rotating its direction (θ_L).

Application Rule

For applied altermagnetic material like in spin-tronics, keeping temperature and strain constant might be essential.

Diagonal Tensor Elements

With the Examples of FeSb_2 and RuO_2

Cobalt-doped FeSb₂

A CASE STUDY IN ALTERMAGNETISM

From Narrow-Gap Semiconductor to Metallic Altermagnet: Optical Fingerprints of Co-Doped FeSb₂

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M. Pinterić,^{1,5} M. van de Loo,² A. Kreyssig² T. Zhang,³ A. E. Böhrer² M. Dressel¹ and M. Wenzel^{1,†}

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²*Experimental Physics IV, Department of Physics and Astronomy,
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(Dated: May 7, 2026)

The realization of bulk metallic altermagnetism has remained elusive despite the growing number of candidate materials. Here, we present evidence that moderate cobalt substitution ($\sim 15\%$) drives the correlated narrow-gap semiconductor FeSb₂ into a metallic altermagnetic state persisting up to room temperature. The infrared optical conductivity reveals low-energy interband transitions near 0.1 eV that emerge upon doping and grow with Co concentration. Density functional theory calculations show that these transitions originate exclusively from altermagnetic spin ordering, with spin split bands (~ 0.2 eV) of non-relativistic origin, together with spin-orbit coupling induced band splitting of the order of ~ 5 meV near the Fermi level. Co substitution further leads to Fano lineshapes and mode mixing in the infrared-active phonons, reflecting enhanced electron-phonon coupling and local inversion symmetry breaking, while leaving the altermagnetic spin symmetry intact. Our results establish carrier-tuned FeSb₂ as a platform for exploring metallic *d*-wave altermagnetism and its coupling to lattice degrees of freedom.

The Parent Compound

FeSb₂ AS A NATURAL ALTERMAGNET CANDIDATE

Orthorhombic Pnnm structure

- **P (Primitive):** Primitive, no additional lattice points except at corners
- **n (1st axis):** n-glide plane perpendicular to the a-axis.
Reflection followed by translation: $t = \frac{b+c}{2}$.
- **n (2nd axis):** n-glide plane perpendicular to the b-axis.
Reflection followed by translation: $t = \frac{a+c}{2}$.
- **m(3rd axis):** A standard Mirror plane perpendicular to the c-axis. This is a simple reflection without any translation.

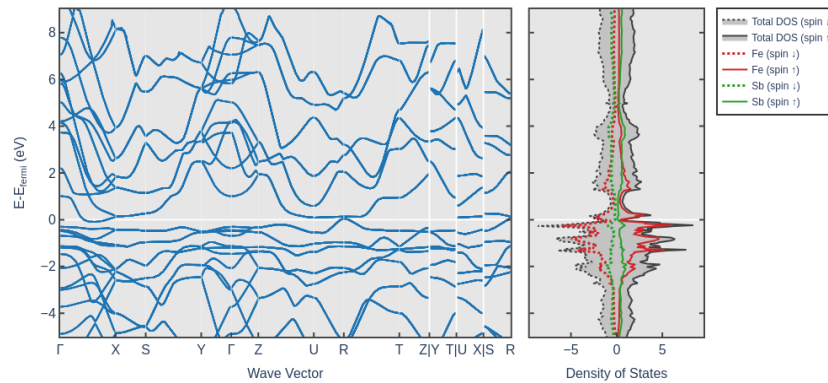
Fe Sublattice is invariant under two-fold rotation + translation $[C_2 \parallel C_{2x}t]$

Makes a **natural altermagnet candidate**

The Parent Compound

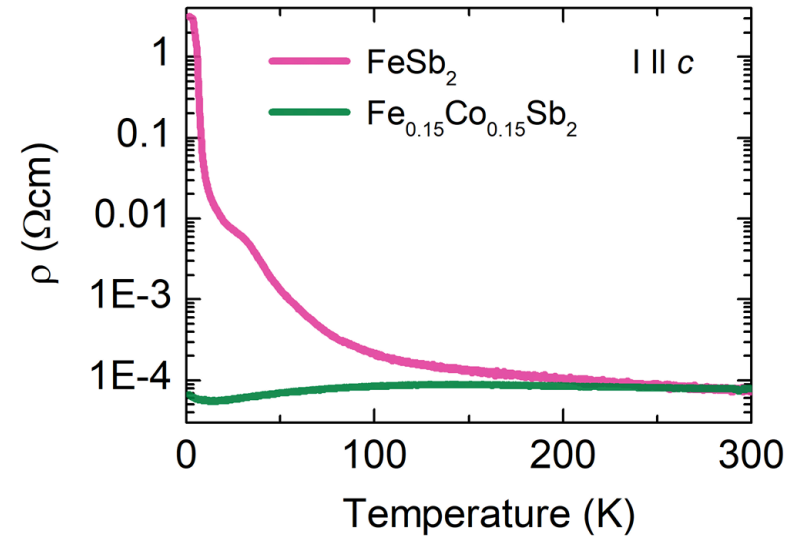
CORRELATED NARROW-GAP SEMICONDUCTOR

BAND STRUCTURE



- Transport gap of approximately 30 meV
- Optical gap of roughly 120 meV

RESISTIVITY

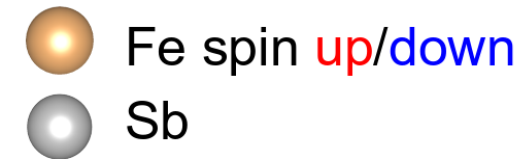


- No long range magnetic order
- Rising resistivity at low temperatures

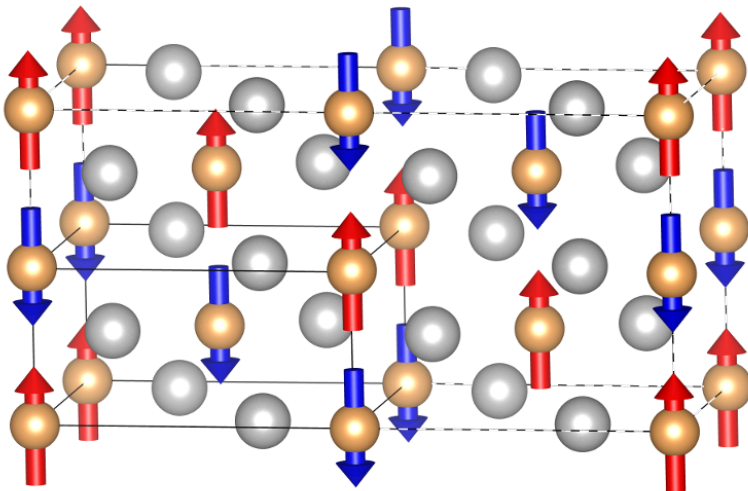
The Parent Compound

CORRELATED NARROW-GAP SEMICONDUCTOR

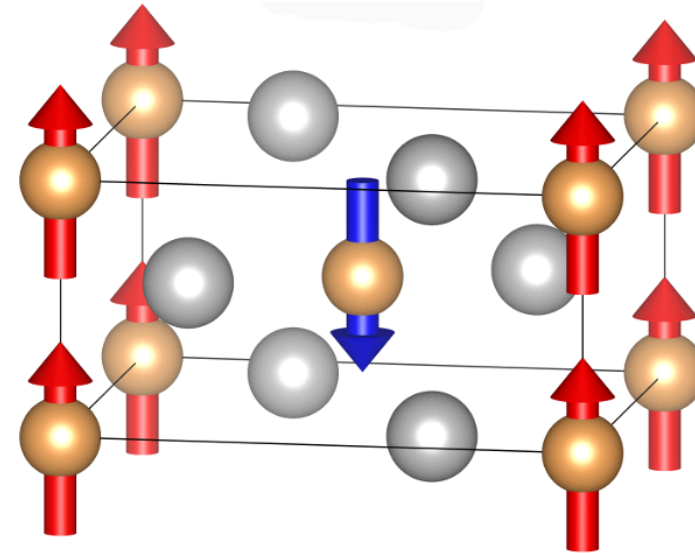
- Transport gap of approximately 30 meV
- Optical gap of roughly 120 meV
- No long range magnetic order
- Rising resistivity at low temperatures



ANTIFERROMAGNETIC (AFME) ORDER



ALTERMAGNETIC (AFMO) ORDER



Cobalt-doped FeSb_2

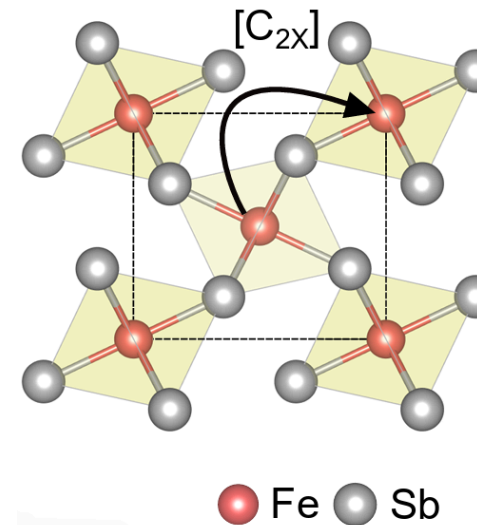
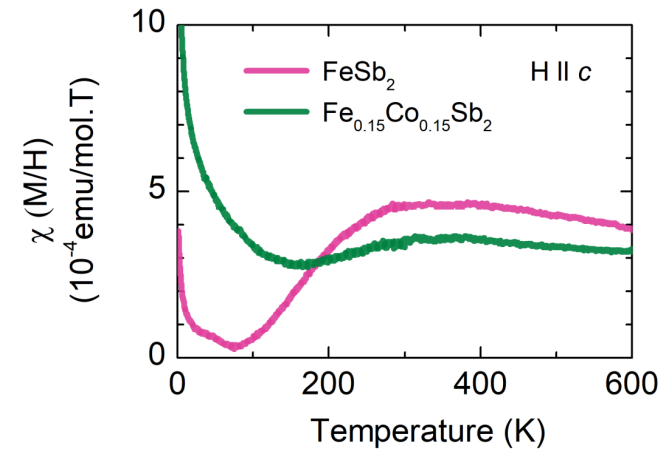
STABILISATION VIA CARRIER DOPING WITH ELECTRONS

Adding electrons/holes

- Adds impurities
- Quenches the spin fluctuations
- Stabilize the AFM order
- Adding holes has similar effect (Cr doping)

Result

- Crystal structure is preserved
- Semiconducting gap is closed
- Persisting up to room temperature

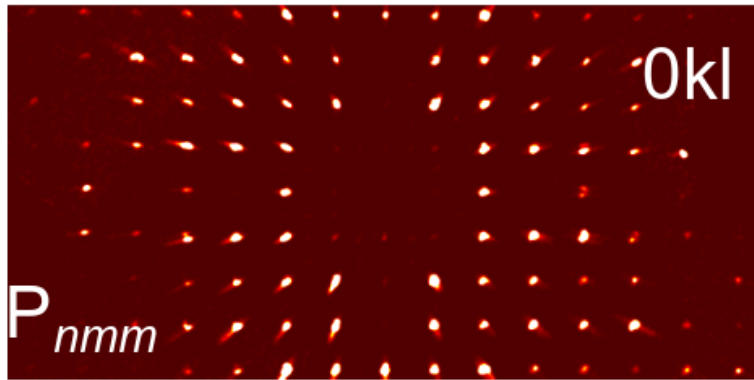


Cobalt-doped FeSb₂

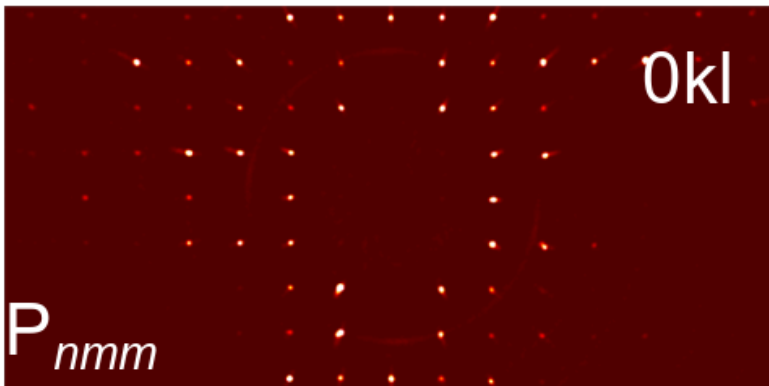
CLOSING THE GAP AND STABILIZING THE ALTERMAGNETIC STATE

X-RAY DIFFRACTION PATTERNS

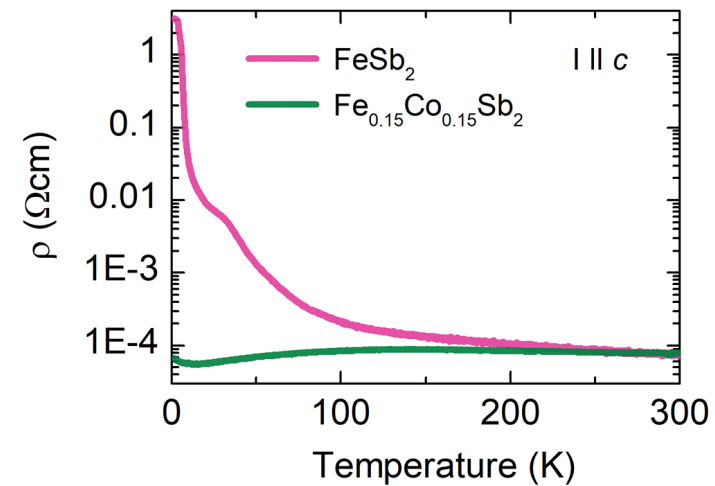
FeSb₂



Fe_{0.85}Co_{0.15}Sb₂



RESISTIVITY



Which magnetization order survived?

→ Compare optical conductivity to DFT calculations!

Preparing the DFT Simulation

SELECTION OF E_{xc} FUNCTIONAL AND MAGNETIC ORDER

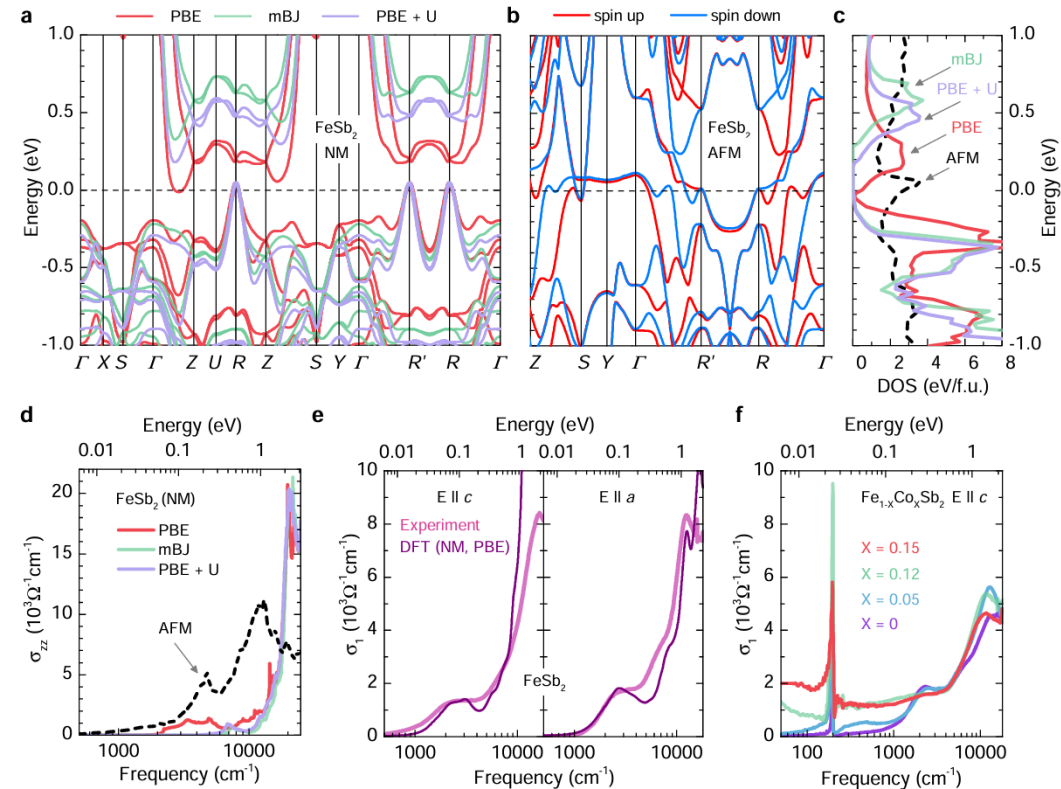
- No matching DOS at E_F
- No matching Band structure
- No matching optical conductivity

Conclusion:

The AFM state is not the ground state of FeSb_2

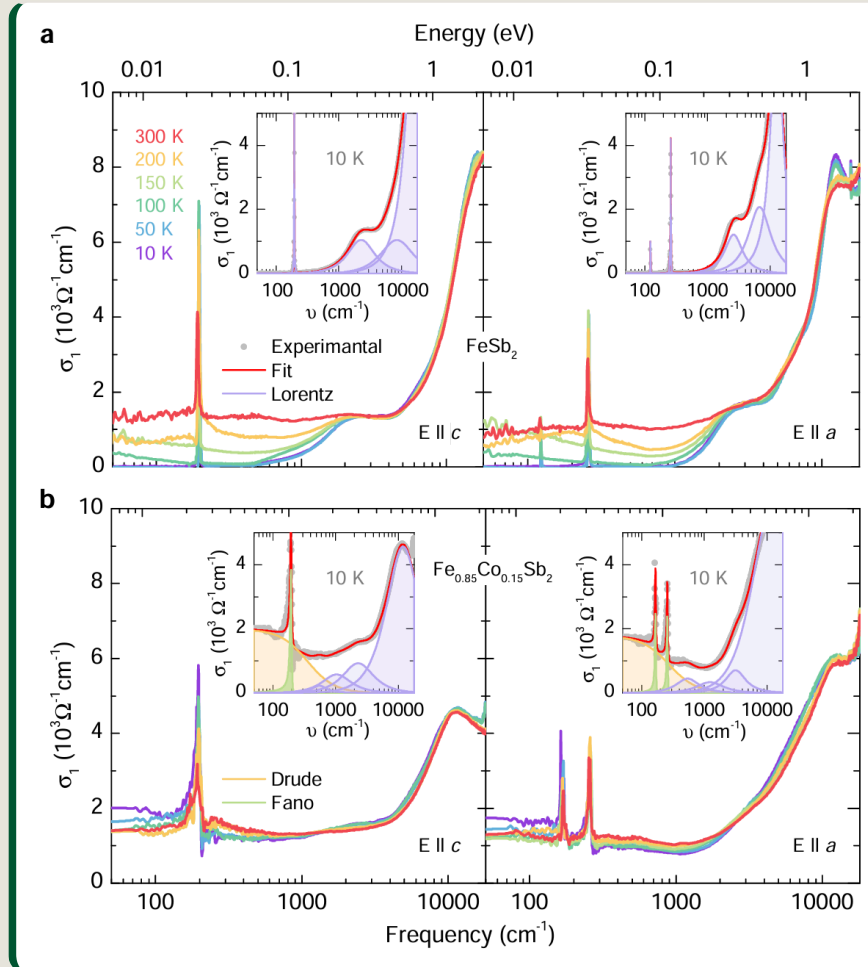
PBE simulates non-magnetic FeSb_2 best.

The underestimation of band gaps aligns with itinerant spin fluctuations.



Identifying the Altermagnetic State

THE DFT COMPARISON (NM VS. AFME VS. AFMO)



Observation

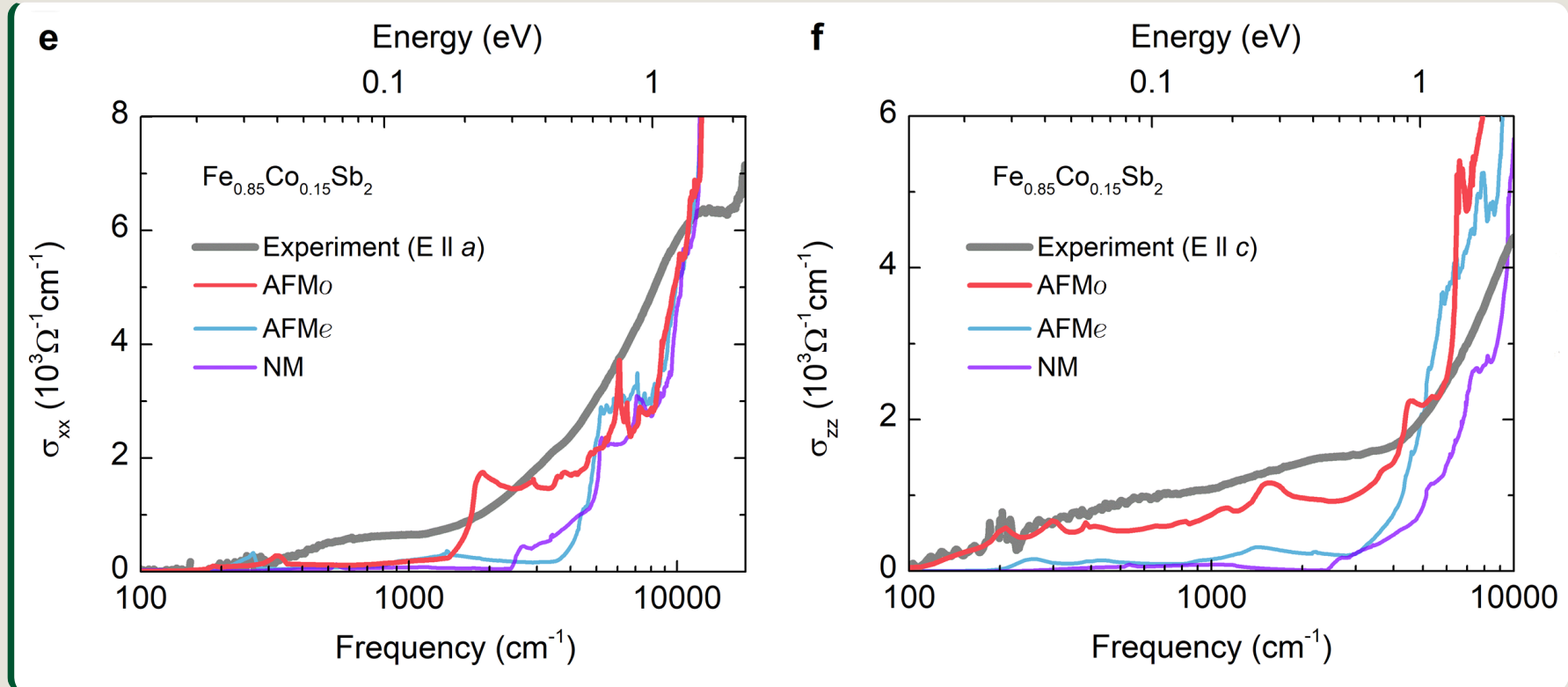
- Metallicity: Drude peak at $\omega = 0$
- New interband transition: peak in doped material at $\sim 800 \text{ cm}^{-1}$

Conclusion

The existence and position are a direct fingerprint of the new band structure. Now, we can compare this fingerprint to our theoretical models.

Identifying the Altermagnetic State

THE DFT COMPARISON (NM VS. AFME VS. AFMO)

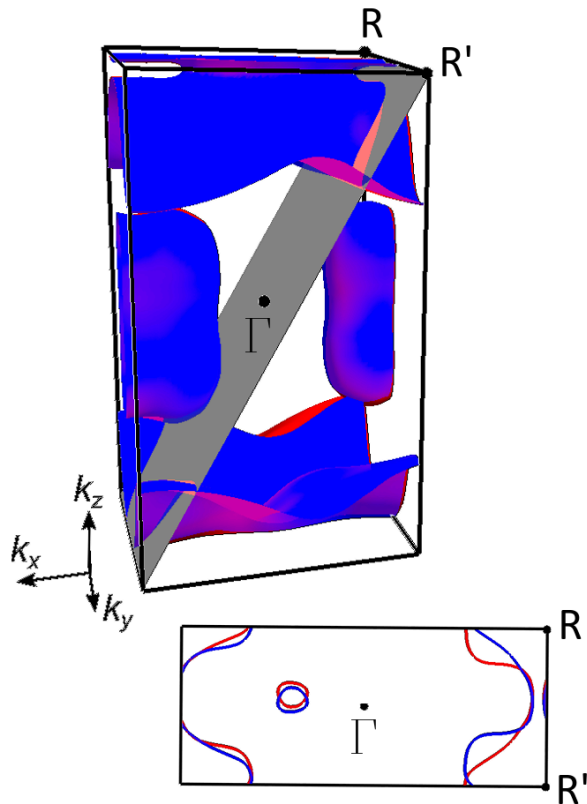


Only the **AFMo** (red line) state shows a good agreement with the experimental data.

Identifying the Altermagnetic State

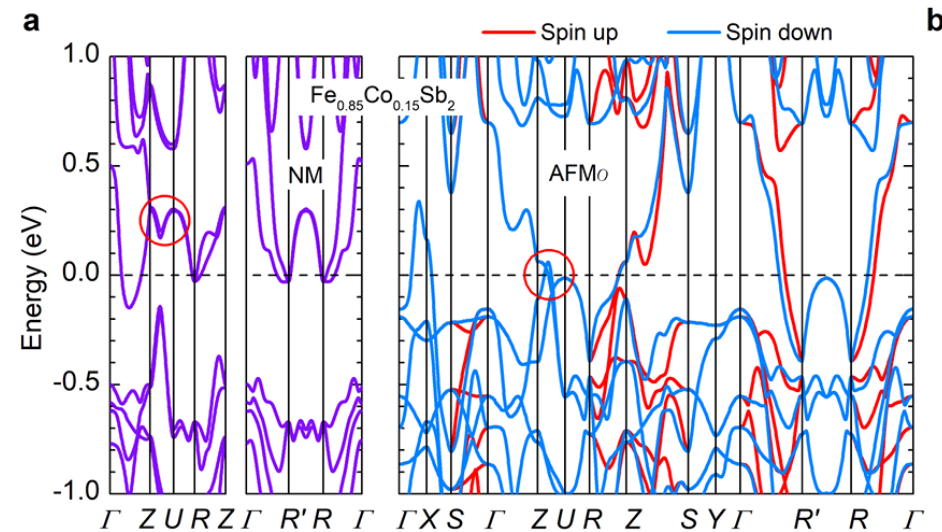
NON-RELATIVISTIC SPIN SPLITTING

Fermi-Surface



- Clear spin-splitting of the Fermi surface
- Alternating spin-polarization across the Brillouin zone

Band-Structure

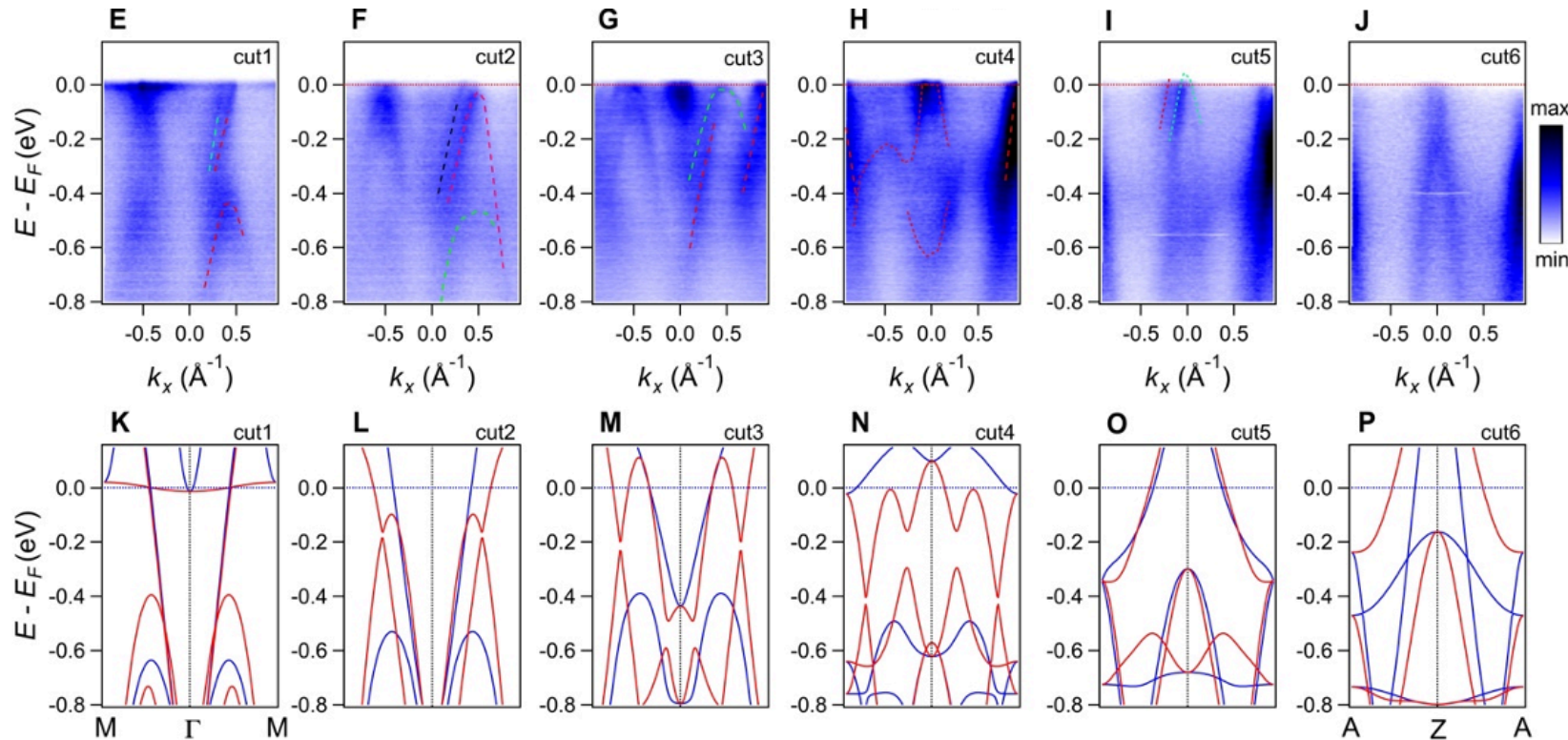


Cobalt doped FeSb_2 is definitely an
Altermagnet

RuO₂ Controversy: A Model Altermagnet?

EARLY PREDICTION OF ALTERMAGNETISM

Surface Probe (ARPES): Evidence for Spin-Splitting



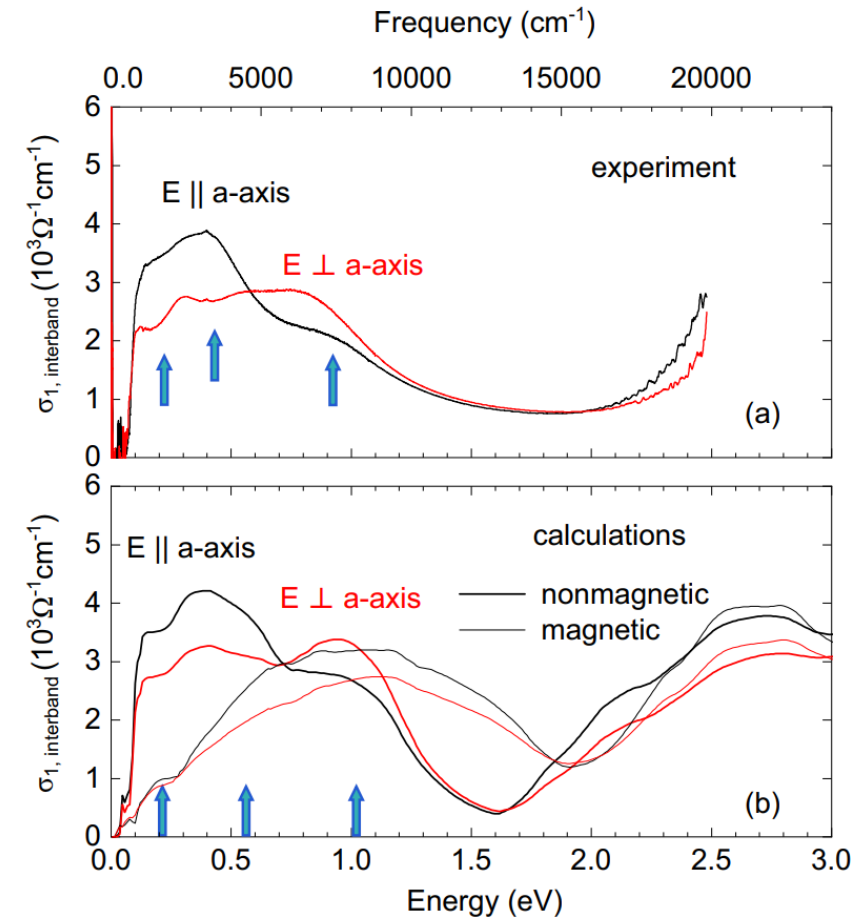
RuO₂ Controversy: A Model Fermi-Liquid?

FERMI-LIQUID BEHAVIOR IN BULK RuO₂

Experiment matches NON-magnetic model

test

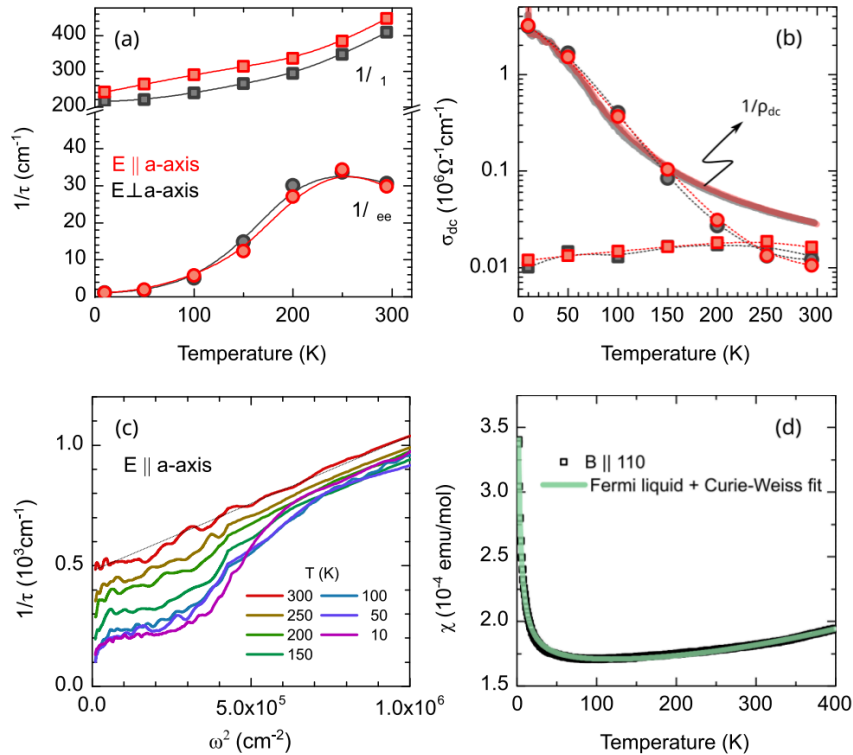
- Sharp **Pauli Edge** around 0.1 eV
- Minimum around 1.5 – 2.0 eV
- Only **non-magnetic** DFT calculation matches



RuO₂ Controversy: A Model Fermi-Liquid!

A PARAMAGNETIC FERMI LIQUID

Other characteristics match fermi liquids too!



Magnetic Susceptibility Data

Figure (d):

- Excellent fit to a model combining Pauli paramagnetism (T^2 dependence)
- Small impurity tail

Phase Angle

Figure (c)/(e):

- Quadratic dependence (ω^2)
- hallmark of a Fermi liquid

Summary

MOKE

- Single ω
- High Sensitivity

MOKKA

- Broad Band
- Low Sensitivity

Mueller

- Model dependent
- Regression Task

Summary

MnTe

Watch out for strain
and temperature

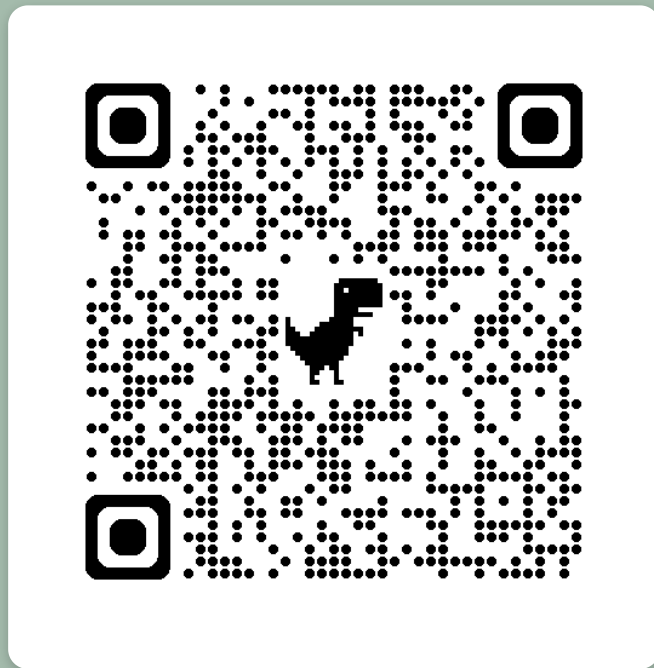
FeSb₂

Confirmed
altermagnetic state in
Co-doped material

RuO₂

Another proof against
altermagnetism

Thank you!



[Link to the presentation](#)

Off-Diagonal Elements: $\alpha - \text{Fe}_2\text{O}_3$

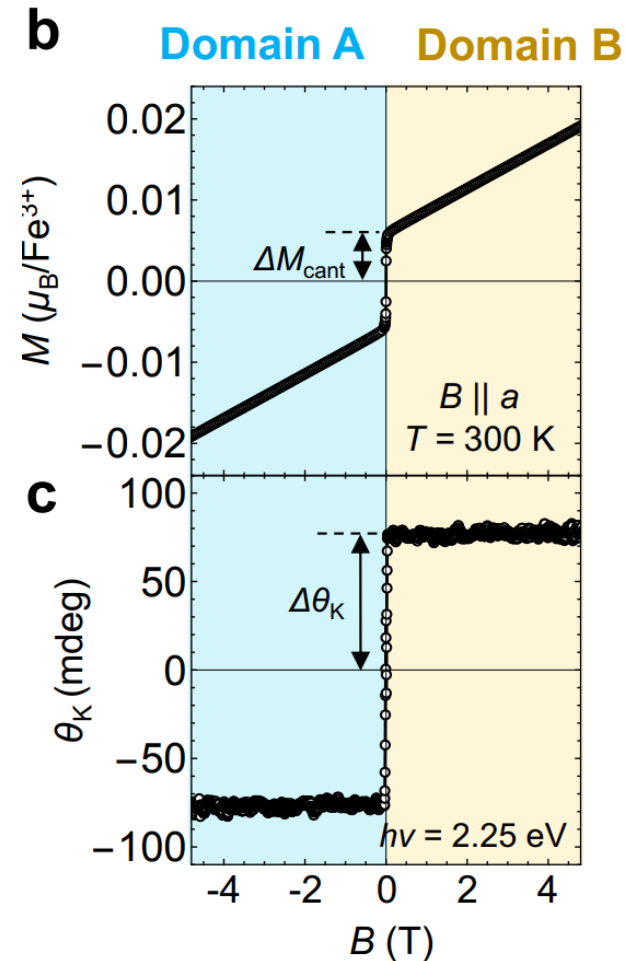
A GIANT MOKE IN A "SIMPLE" ANTIFERROMAGNET

The Candidate: Hematite

- $\alpha - \text{Fe}_2\text{O}_3$ is a **collinear** antiferromagnetic insulator.
- With zero net magnetization ($M=0$), it was long considered "invisible" to MOKE.
- But crystal structure allows for the breaking of $\mathcal{TtS}$ symmetry.

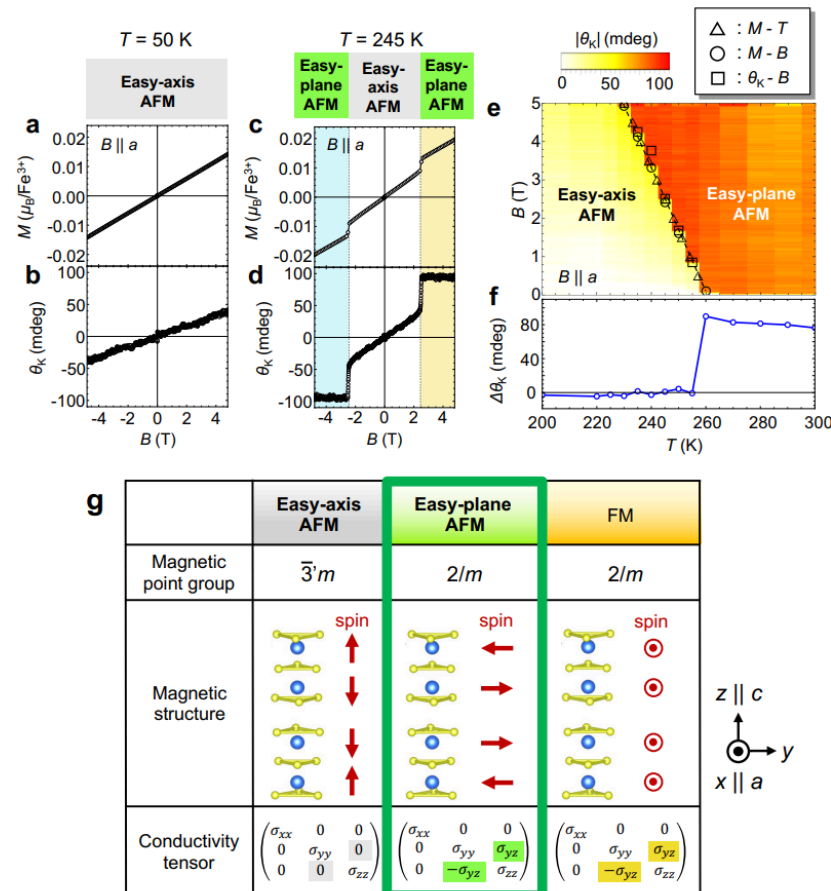
The Experimental Surprise

- A **giant spontaneous Kerr effect** ($\theta_K \approx 80$ mdeg) is observed.
- The tiny net magnetization ($\Delta M \approx 0.006\mu_B/\text{Fe}$) is far too small to explain this signal.



The Answer is Symmetry

THE MORIN TRANSITION AS AN "ON/OFF" SWITCH



Easy-Plane Phase ($T > 260$ K)

- Spins lie in the basal plane.
- The magnetic point group ($2/m$) **allows** for non-zero off-diagonal elements (σ_{yz}).
- This symmetry creates a "fictitious magnetic field" generating the MOKE signal.

Easy-Axis Phase ($T < 260$ K)

- Spins reorient along the c-axis during the "Morin transition".
- The new magnetic point group ($\bar{3}'m$) **forbids** any off-diagonal elements.
- The spontaneous Kerr effect vanishes, proving it is tied to the specific AFM order.

DFT Confirms the Microscopic Origin

THEORY VS. EXPERIMENT FOR THE EASY-PLANE PHASE

Decisive Test

- First-principles calculations were performed for the easy-plane AFM state.
- Crucially, the simulation was run assuming **zero net magnetization ($M=0$)**.

Perfect Match

- The simulation perfectly reproduces the experimental Kerr rotation (θ_K) and ellipticity (η_K).
- This includes the spectral shape, peak position (~ 2.5 eV), and absolute magnitude.
- This provides conclusive evidence that the giant MOKE is an intrinsic property of the TtS-broken AFM order itself.

